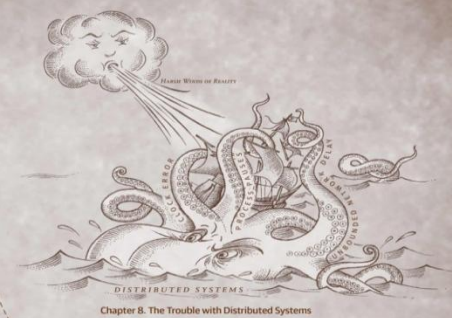




Big Data Systems Introduction

Thorsten Papenbrock

Philipps-University of Marburg

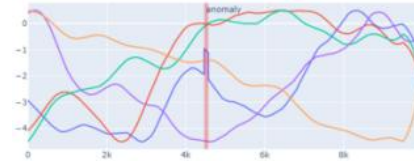
Big Data Analytics Group

• Research

- Time Series Analytics (Zeitreihenanalyse)
(Anomaly and Pattern Detection, Time Series Prediction, ...)
- Data Profiling (Metadatenanalyse)
(Dependency Discovery, Metadata Application, ...)
- Scalable Computing (Lastverteilung)
(Big Data Systems, Reactive Algorithms, Microservices, ...)
- Data Management (Informationssysteme)
(Data Cleaning, Integration, Transformation, Storage, ...)

• Teaching

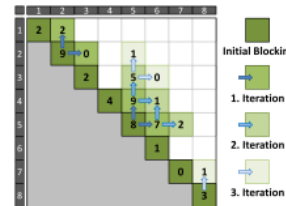
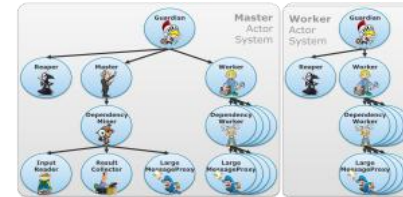
- Big Data Systems (WiSe, VL, Master)
- Data Integration (SoSe, VL, Master)
- Database Systems (SoSe, VL, Bachelor)



Inter-table dependencies		relationships				key candidates		
Planet Name < Planet		Planet < Confirmed Moons				Planet < Atmosphere		
Name	Type	Equatorial diameter	Mass	Orbital radius	Orbital period	Rotation period	Confirmed moons	Rings
Mercury	Terrestrial	0.382	0.06	0.47	0.24	58.64	0	no
Venus	Terrestrial	0.949	0.82	0.72	0.62	~243.02	0	no
Earth	Terrestrial	1.000	1.00	1.00	1.00	1.00	1	no
Mars	Terrestrial	0.532	0.11	1.52	1.88	1.03	2	no
Jupiter	Giant	11.209	317.80	5.20	11.86	0.41	67	yes
Saturn	Giant	9.449	95.2	9.54	29.46	0.43	62	yes
Uranus	Giant	4.007	14.6	19.22	84.01	-0.72	27	yes
Neptune	Giant	3.883	17.2	30.06	164.8	0.67	14	yes

(Diameter < Mass > 0) (orbital radius < orbital period) (Atmosphere < Rings)

hidden constraints order dependencies intra-table dependencies



Prof. Thorsten
Papenbrock



Ilnaz
Tayebi



Florian
Siepe



Sebastian
Schmidl



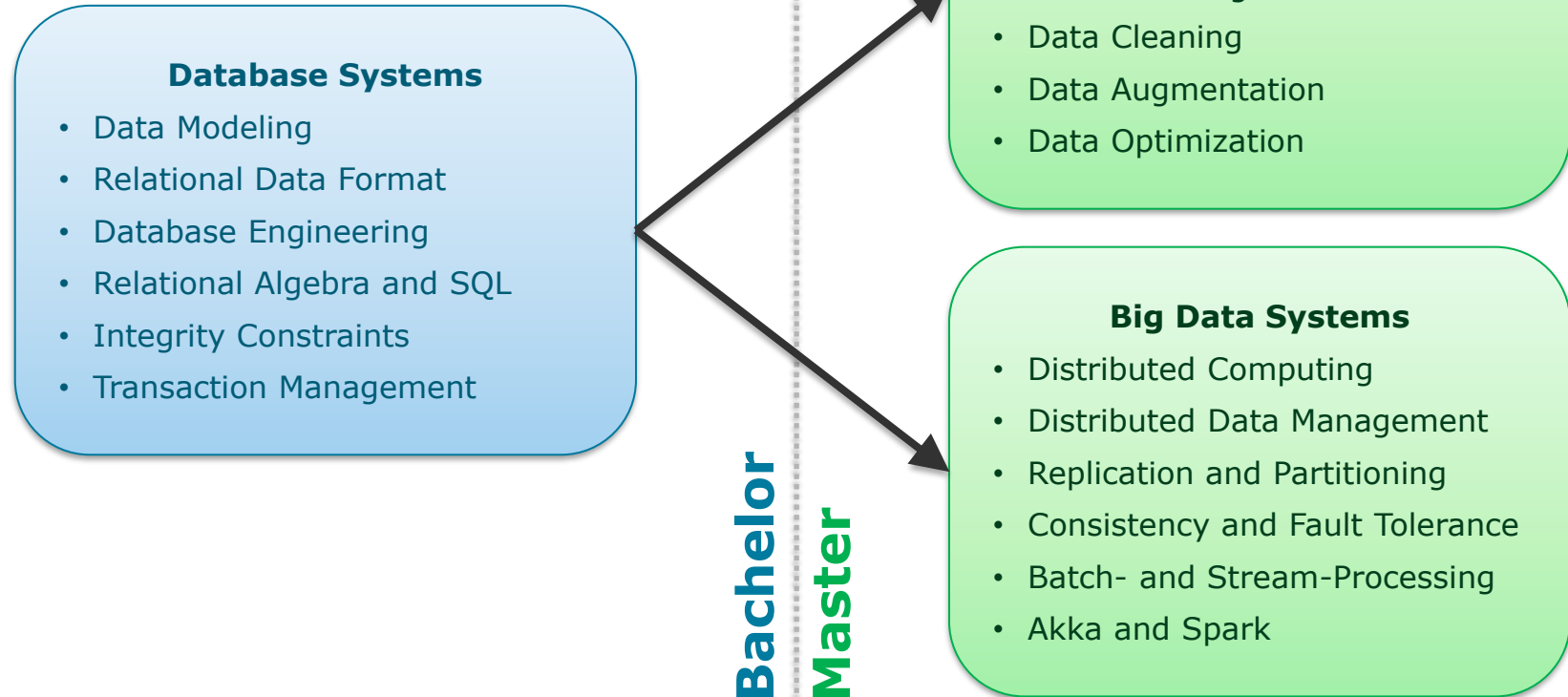
Marcian
Seeger



Thomas
Prinz

Philipps-University of Marburg

Big Data Analytics Group



The background of the slide is a scenic photograph of a town, likely Marburg, with a prominent church spire visible on the right. The town is nestled in a valley, surrounded by green hills and forests. In the foreground, a person with long blonde hair, wearing a red shirt, is sitting on a rocky ledge, looking out over the town. The sky is blue with some light clouds.

Big Data Systems Audience

Language: Only German, English/German, English?

Background: Uni Marburg, International?

Level: Master, Bachelor, Diplom ...?

Semester: 1, 2, 3, 4, 5, 6, 6+?

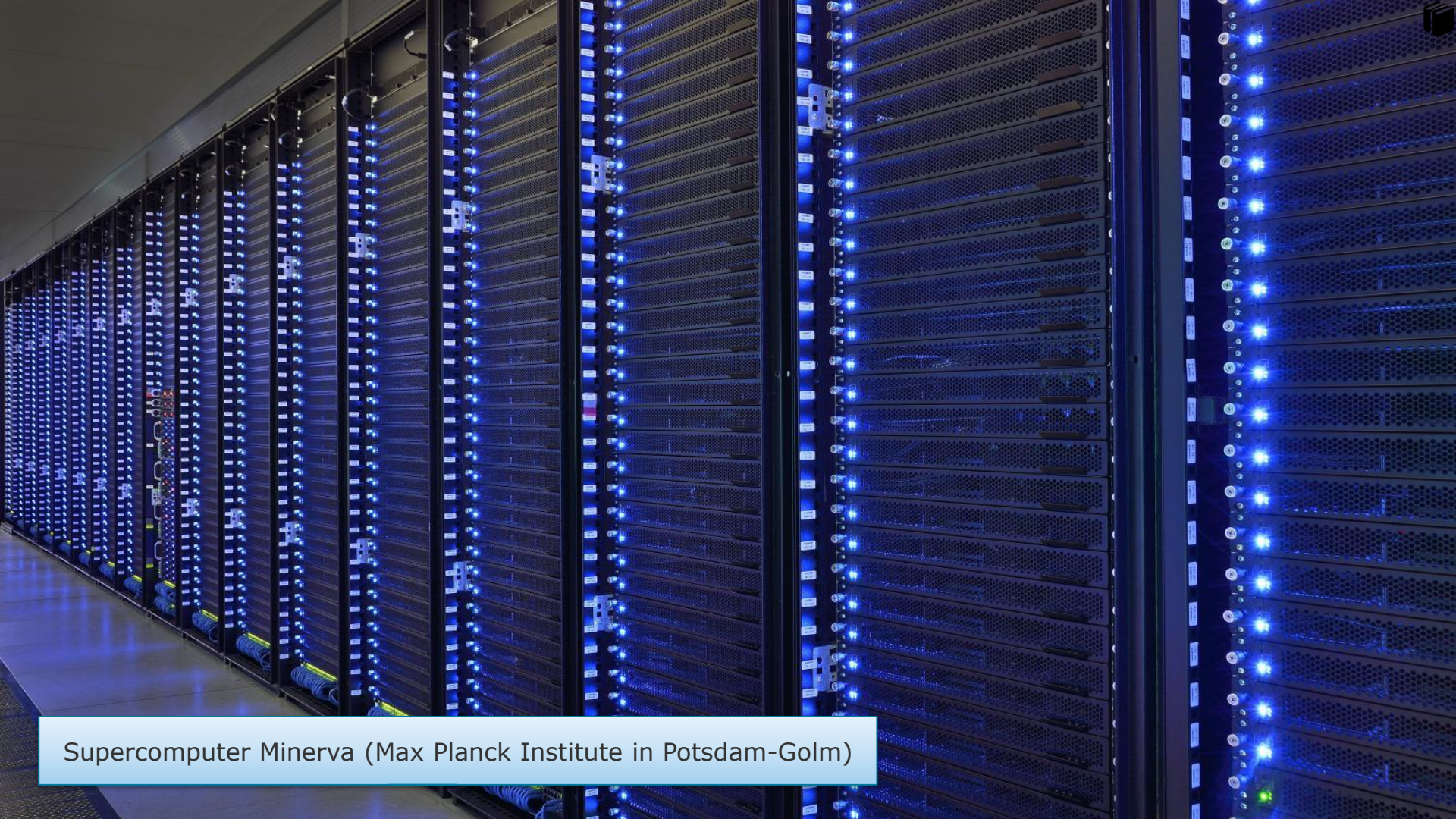
Degree Programs: Computer Science, Data Science, Math ...?

Knowledge: Database Systems, Data Integration, Parallel Computing, Distributed Computing, Internet & Web Technologies, Big Data Systems ...?

Introduction

- **Distributed Systems**
- Lecture Organization
- Motivation “Distributed”
- Motivation “Data”
- Motivation “Management”

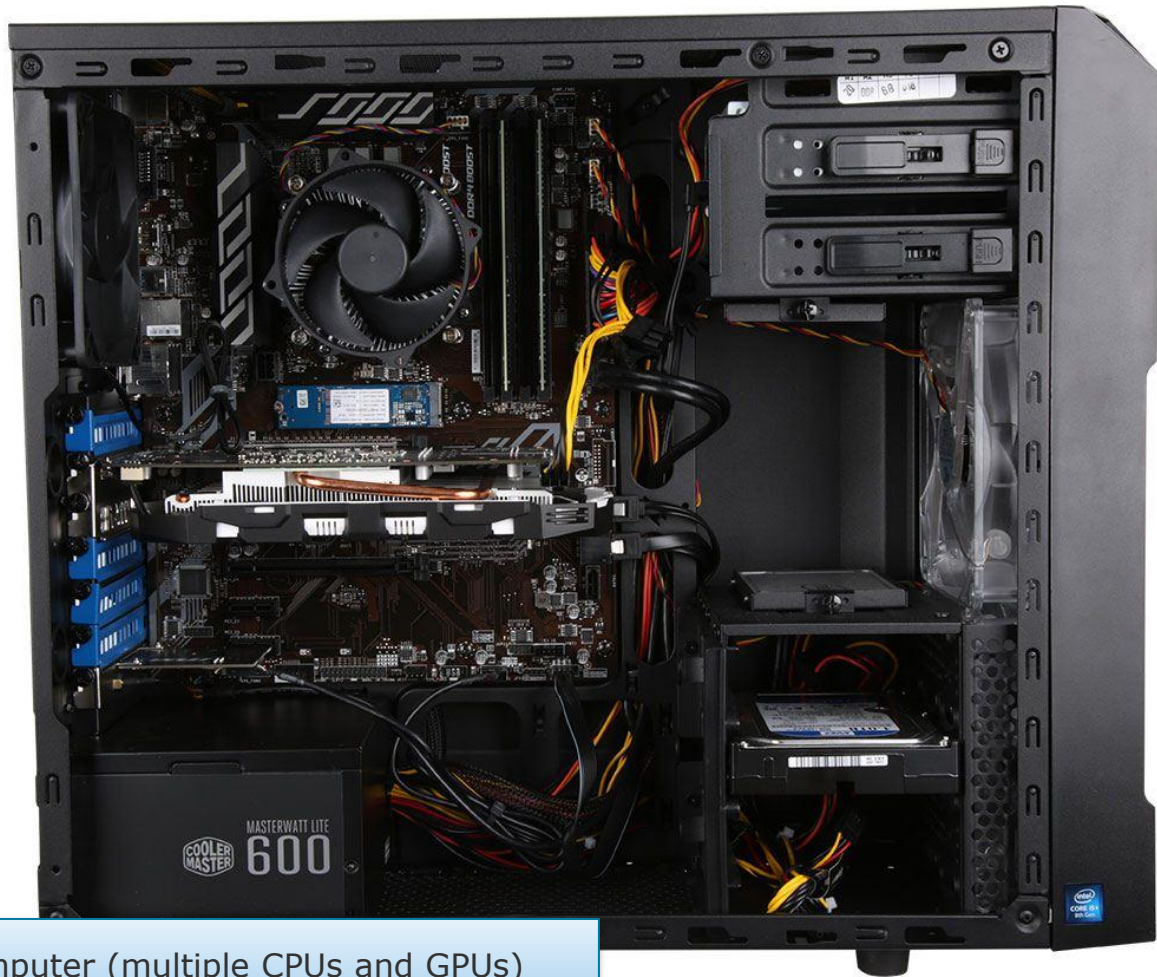




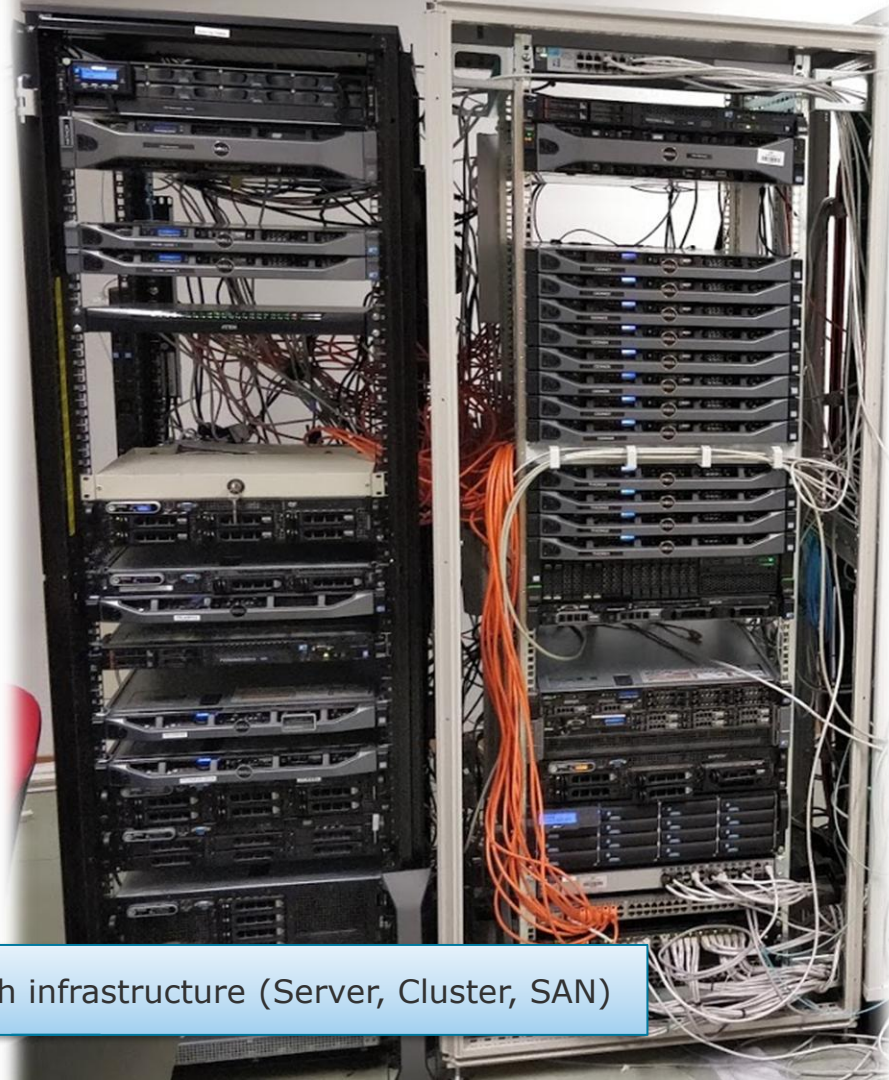
Supercomputer Minerva (Max Planck Institute in Potsdam-Golm)



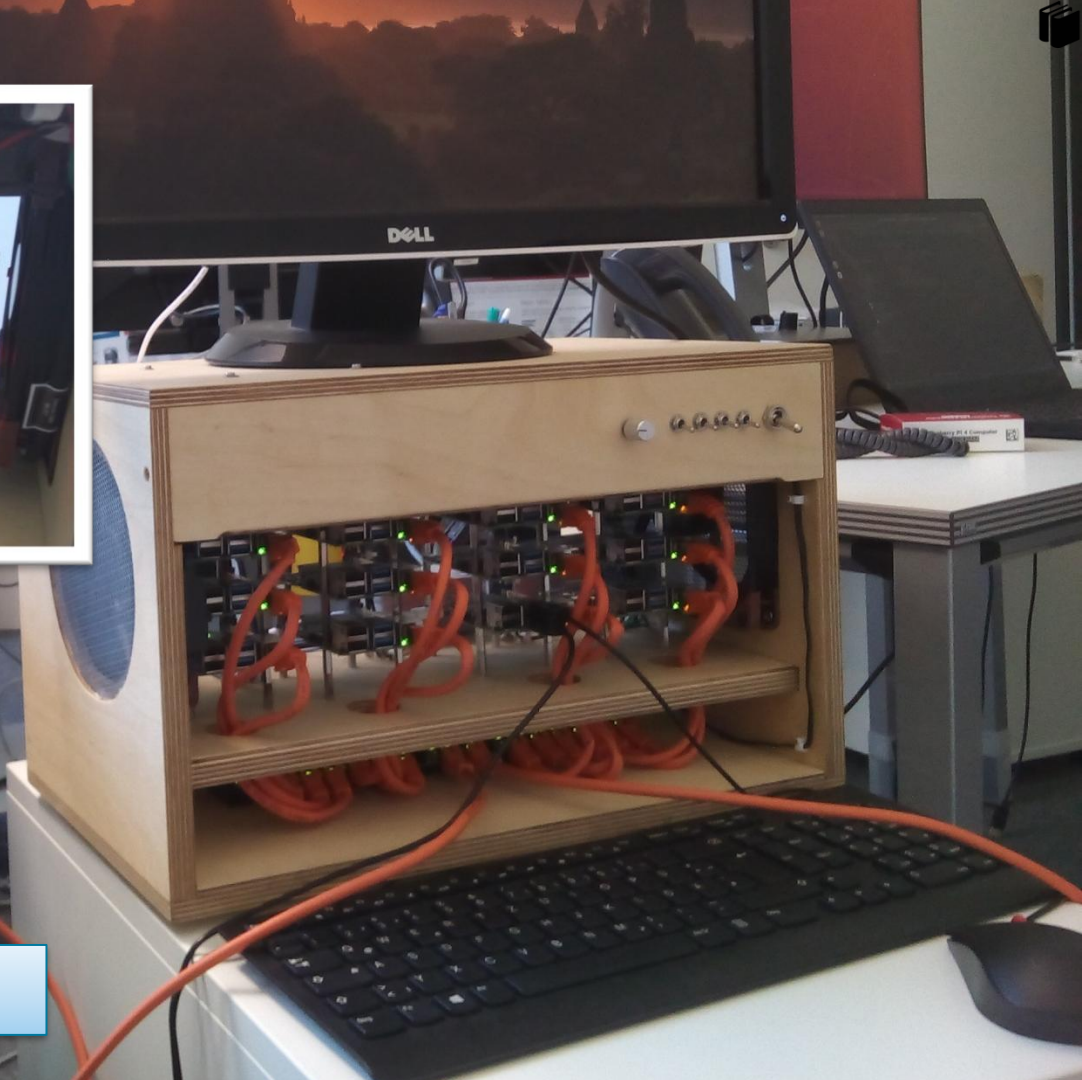
Commodity Hardware Cluster ~2010



Desktop Computer (multiple CPUs and GPUs)



Typical academic research infrastructure (Server, Cluster, SAN)



PI Cluster (12 Raspberry PI 4)

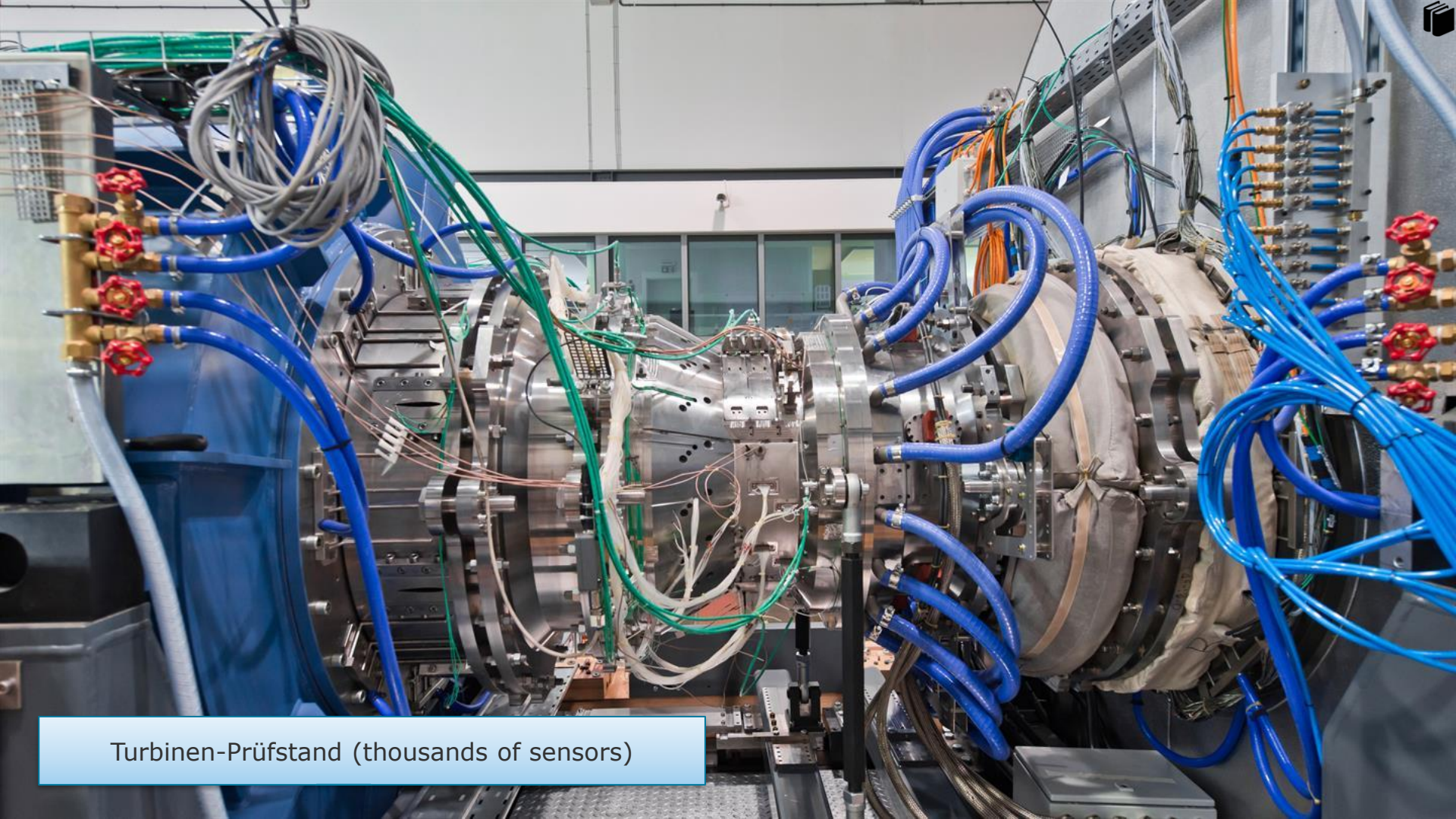


DreamHack (12,000-computer LAN party)



Boeing 747 (thousands of computers)





Turbinen-Prüfstand (thousands of sensors)



Startpage.com



The world's **most private** search engine

Startpage (search engine backed by other search engines)



Outdated Data

Concurrency

Consensus

**Lost & Invalid
Messages**

Termination

Introduction

- Distributed Systems
- **Lecture Organization**
- Motivation “Distributed”
- Motivation “Data”
- Motivation “Management”



Lecture Organization

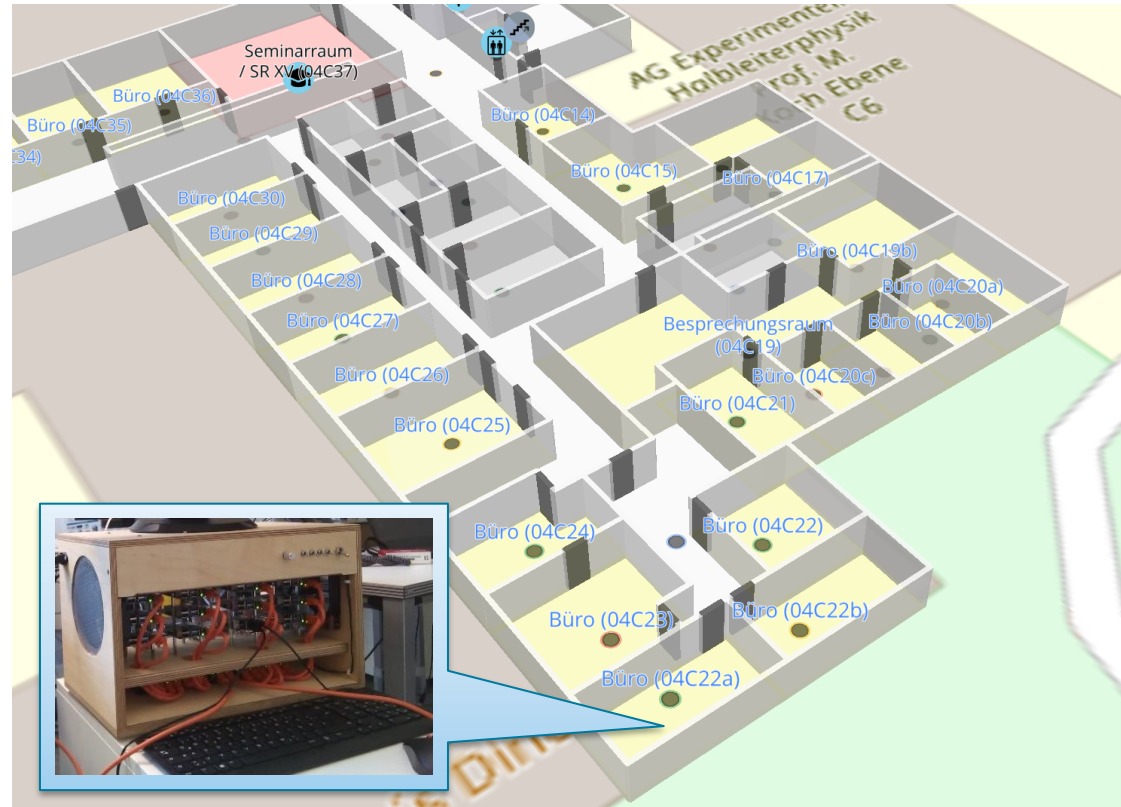
Lecture and Exercise Times

Lecture

- For master students:
"Vertiefungsmodul"
- In module catalog:
"Distributed Data Management"
- 9 credit points, 6 SWS
- Mondays 10:00 – 12:00
Tuesdays 08:00 – 10:00

Exercises

- Teaching:
interleaved with lectures
- Experimentation:
Wednesdays 08:00 – 10:00
04C22a



Lecture Organization

Prerequisites

Prerequisites

- For participation:
 - Knowledge in **database systems**
 - Knowledge in **object oriented programming**
- For exam:
 - Passing of the **Akka and Spark assignments**

Exam

- Written exam
 - 23.02.2026
 - 17.03.2026

Slides

- On the Ilias course page

Big Data Systems

Introduction

Thorsten Papenbrock

Slide **18**



Inhalt



Virtual Meeting Room

BigBlueButton streams of lectures and exercises; BigBlueButton



Forum

A place to ask questions, spark discussions and form exercise groups
Beiträge (Ungelesen): 0 (0)

Lecture Slides

Slides of all completed lectures and exercises



Homework Data

Data and resources for the practical exercises



Exercise Exams

Exercises and their solutions of past DDM iterations



Literature

Reference publications and books for the topics discussed in the lecture



Coding Assignments

Submit your GitHub repository
Lernfortschritt: 0/0

Lecture Recordings

Recordings of the BigBlueButton streams of the lecture

Hybrid lecture: in-person & stream

Questions and team building!

Submit your GitHub repositories!

Neigkeiten

Die Datei wurde hinzugefügt.

Datei: BD5_Re_Exam_WS2023_Musterloes...

Termin: Heute, 11:06

Die Datei wurde hinzugefügt.

Datei: BD5_Re_Exam_WS2023_2.pdf

Termin: Heute, 11:06

Die Datei wurde hinzugefügt.

Datei: ExamDDM-5521 solution.pdf

Termin: Heute, 11:06

Die Datei wurde hinzugefügt.

Datei: ExamDDM-WS1920 solution.pdf

Termin: Heute, 11:06

Die Datei wurde hinzugefügt.

Datei: ExamDDA-WS1718.pdf

Termin: Heute, 11:06

Kalender

So	Mo	Di	Mi
			1
5	6	7	8
12	13	14	15
19	20	21	22
26	27	28	29

Lecture Organization

Marvin Details

Big Data Systems ▾

Grunddaten

Verantwortliche/-r

• [Prof. Dr. Thorsten Papenbrock](#)

Termine

 [Daten für iCalendar \(ics\) exportieren](#)  [Einzeltermine anzeigen](#)  [Details einblenden](#)

Änderungen	Rhythmus	Wochentag	Von - Bis	Ausfalltermin	Startdatum - Enddatum	Erw. Tn.	Bemerkung	Durchführende/- r	Raum
	wöchentlich	Mo	10:00 - 12:00	22.12.2025 29.12.2025 05.01.2026	13.10.2025 - 09.02.2026				 04A30 (HS IV A4) (Hans-Meerwein-Straße 6, Institutsgebäude (H. 04))
	wöchentlich	Di	08:00 - 10:00	23.12.2025 30.12.2025 06.01.2026	14.10.2025 - 10.02.2026				 04A30 (HS IV A4) (Hans-Meerwein-Straße 6, Institutsgebäude (H. 04))

Lecture Organization

Feedback

Question at any time please!

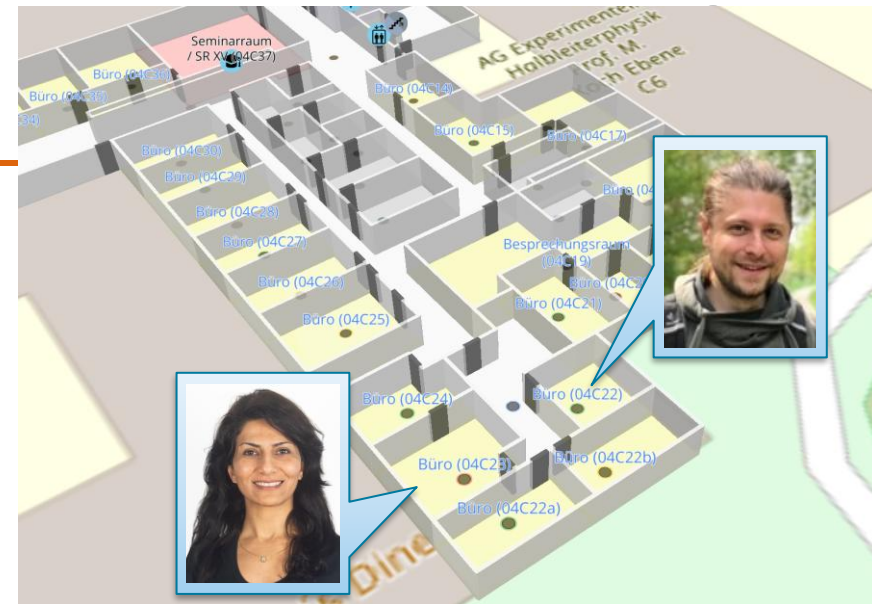
- During lectures
- Visit us: 04C22 (Aufgang C, Ebene 4)
- Email:
 - papenbrock@informatik.uni-marburg.de

Feedback is welcome!

- Lecture improvements
- Informational material
- Organization

Official evaluation

- At the end of this semester
- ... too late for important feedback!



Big Data Systems

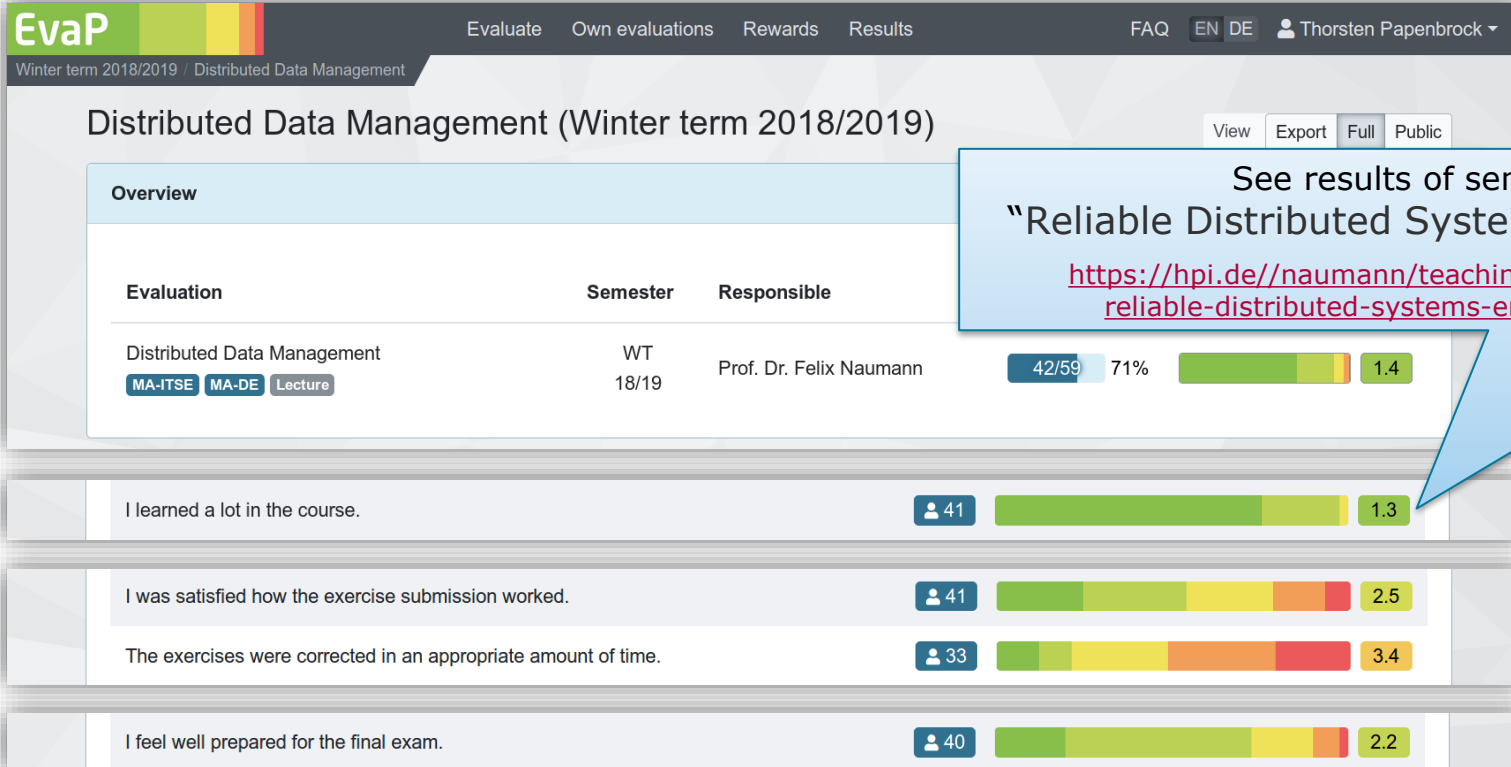
Introduction

Thorsten Papenbrock

Slide **21**

Lecture Organization

Feedback



"A lot of content!"

Big Data Systems
Introduction

Thorsten Papenbrock
Slide 22

Lecture Organization

Outline

1. Introduction
2. Foundations
3. Hands-On: Akka
4. Encoding
5. Communication
6. Data Models and Query Languages
7. Storage and Retrieval
8. Replication
9. Partitioning



10. Distributed Systems
11. Consistency and Consensus
12. Transactions
13. Batch Processing
14. Hands-On: Spark
15. Stream Processing
16. Distributed DBMSs
17. Distributed Query Optimization
18. Lecture Summary and Exam Preparation



Big Data Systems

Introduction

Lecture Organization

Outline

1. Introduction
2. Foundations
3. Hands-On: Akka
4. Encoding
5. Communication
6. Data Models and Query Languages
7. Storage and Retrieval
8. Replication
9. Partitioning



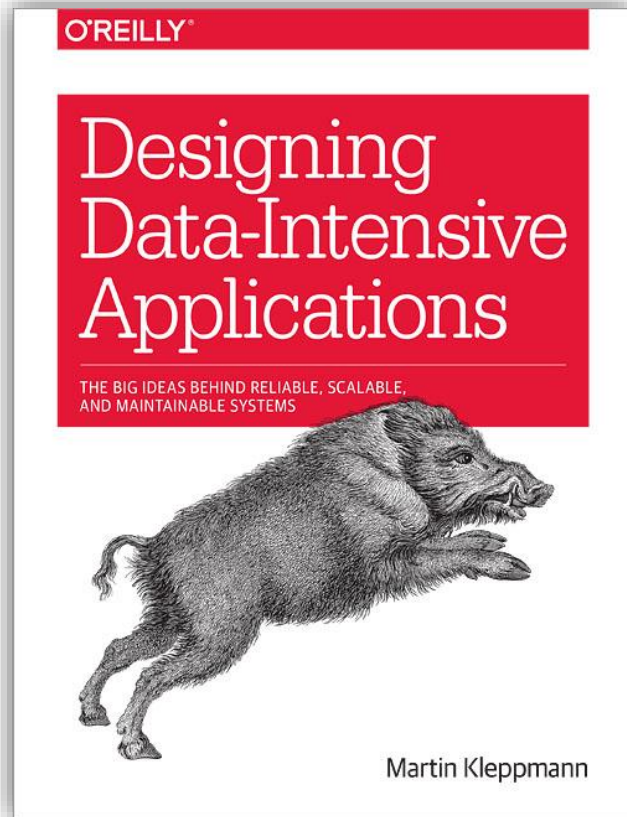
10. Distributed Systems
11. Consistency and Consensus
12. Transactions
13. Batch Processing
14. Hands-On: Spark
15. Stream Processing
16. Distributed DBMSs
17. Distributed Query Optimization
18. Lecture Summary and Exam Preparation



Big Data Systems

Introduction

ThorstenPapenbrock
Slide **24**



Designing Data-Intensive Applications

- Author: Martin Kleppmann
- Date: March 2017
- Publisher: O'Reilly Media, Inc
- ISBN: 978-1-449-37332-0
- References:
<https://github.com/ept/ddia-references>

Scope for this lecture

- Distributed and parallel systems
- Big data storage
- Batch and stream processing

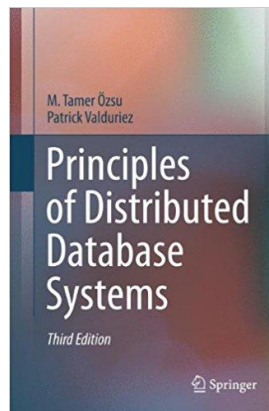
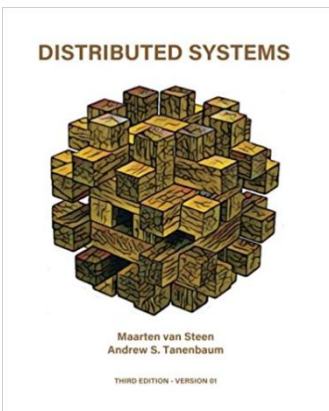
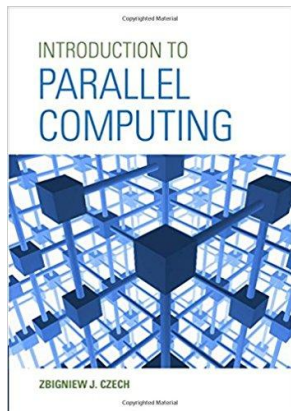
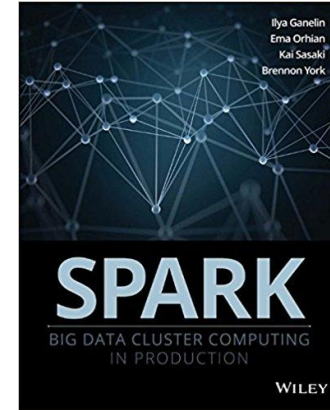
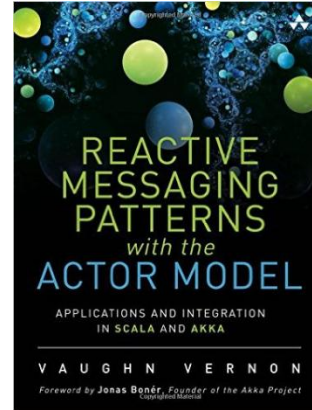
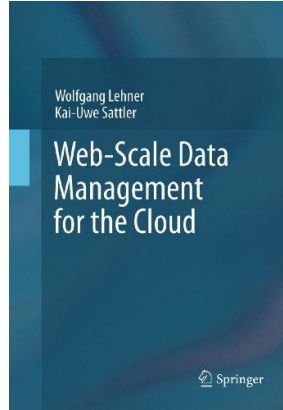
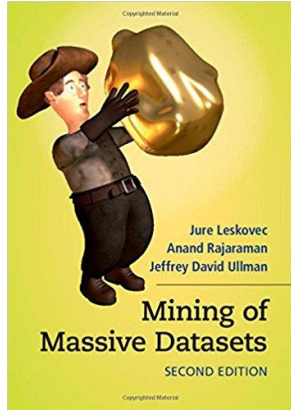
Big Data Systems

Introduction

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Slide **25**

Lecture Organization

Literature: Further Reading



And Web-links that
are given on the slides
during the lecture.

Big Data Systems

Introduction

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Slide **26**

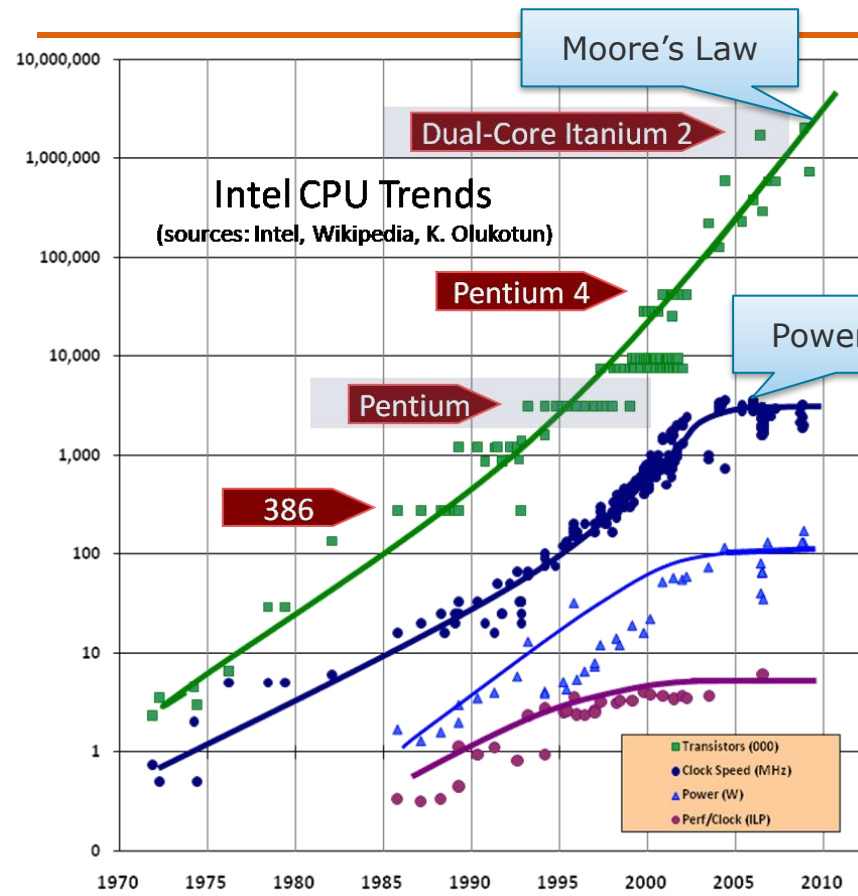
Introduction

- Distributed Systems
- Lecture Organization
- **Motivation “Distributed”**
- Motivation “Data”
- Motivation “Management”



Motivation: "Distributed"

Paradigm Shift in Software-Writing



The free lunch is over!

- Clock speeds stall
- Transistor numbers still increase
 - Cores in CPUs/GPUs
 - CPUs/GPUs in compute nodes, compute nodes in clusters
- Paradigm Shift:
 - Earlier: optimize code for a single thread
 - Now: solve tasks in parallel

Distributed computing

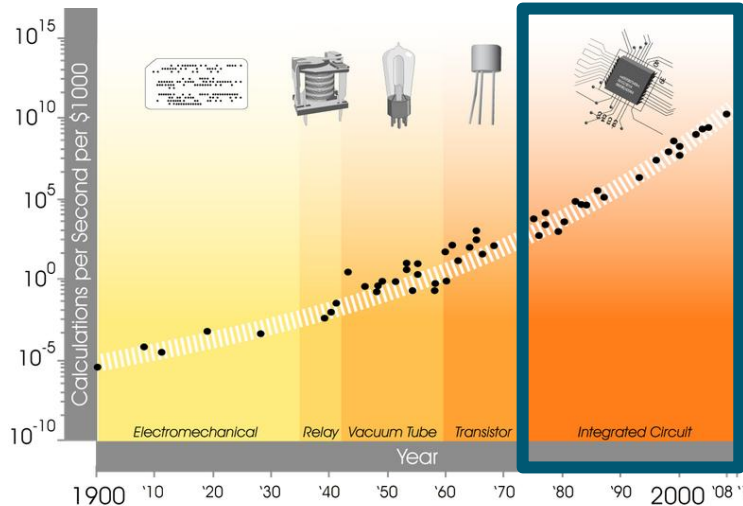
"Distribution of work on (potentially) physically isolated compute nodes"

Philipps Universität
Marburg

Hyperscale: With clusters of **distributed machines**, we can already build systems with any number of transistors!
(don't even need to wait for a new processors)

The graph illustrates the exponential growth of transistors in integrated circuits over time. The y-axis represents the transistor count on a logarithmic scale, ranging from 1,000 to 50,000,000,000. The x-axis represents the year, from 1970 to 2018. The data points show a clear upward trend, with the transistor count increasing by approximately two orders of magnitude every two years. Key milestones include the Intel 4004 (1971, 2,300 transistors), the Intel 8086 (1982, 290,000 transistors), the Intel Pentium (1992, 3.1 million transistors), and the Intel Core i7 (2008, 731 million transistors). The graph also shows the emergence of multi-core processors, such as the Intel Core i7 (2008, 731 million transistors) and the Intel Xeon Phi (2015, 5.3 billion transistors).

Processor	Year	Transistor Count
Intel 4004	1971	2,300
TMS 1000	1971	6,000
Intel 8008	1972	6,000
Zilog Z80	1974	6,000
Intel 8080	1974	6,000
Motorola 6800	1974	6,000
Intel 8085	1976	6,000
Intel 8086	1978	290,000
Motorola 68000	1978	290,000
Intel 8088	1982	290,000
Intel 80286	1982	290,000
Motorola 68020	1982	290,000
Intel 80386	1985	2,750,000
TI Explorer's 32-bit laptop machine chip	1985	2,750,000
Intel 80486	1989	12,000,000
ARM 3	1989	12,000,000
DEC VRL Multi-11	1989	12,000,000
ARM 6	1992	275,000
ARM 700	1992	275,000
ARM 10	1992	275,000
ARM 11	1992	275,000
ARM 12	1992	275,000
ARM 13	1992	275,000
ARM 14	1992	275,000
ARM 15	1992	275,000
ARM 16	1992	275,000
ARM 17	1992	275,000
ARM 18	1992	275,000
ARM 19	1992	275,000
ARM 20	1992	275,000
ARM 21	1992	275,000
ARM 22	1992	275,000
ARM 23	1992	275,000
ARM 24	1992	275,000
ARM 25	1992	275,000
ARM 26	1992	275,000
ARM 27	1992	275,000
ARM 28	1992	275,000
ARM 29	1992	275,000
ARM 30	1992	275,000
ARM 31	1992	275,000
ARM 32	1992	275,000
ARM 33	1992	275,000
ARM 34	1992	275,000
ARM 35	1992	275,000
ARM 36	1992	275,000
ARM 37	1992	275,000
ARM 38	1992	275,000
ARM 39	1992	275,000
ARM 40	1992	275,000
ARM 41	1992	275,000
ARM 42	1992	275,000
ARM 43	1992	275,000
ARM 44	1992	275,000
ARM 45	1992	275,000
ARM 46	1992	275,000
ARM 47	1992	275,000
ARM 48	1992	275,000
ARM 49	1992	275,000
ARM 50	1992	275,000
ARM 51	1992	275,000
ARM 52	1992	275,000
ARM 53	1992	275,000
ARM 54	1992	275,000
ARM 55	1992	275,000
ARM 56	1992	275,000
ARM 57	1992	275,000
ARM 58	1992	275,000
ARM 59	1992	275,000
ARM 60	1992	275,000
ARM 61	1992	275,000
ARM 62	1992	275,000
ARM 63	1992	275,000
ARM 64	1992	275,000
ARM 65	1992	275,000
ARM 66	1992	275,000
ARM 67	1992	275,000
ARM 68	1992	275,000
ARM 69	1992	275,000
ARM 70	1992	275,000
ARM 71	1992	275,000
ARM 72	1992	275,000
ARM 73	1992	275,000
ARM 74	1992	275,000
ARM 75	1992	275,000
ARM 76	1992	275,000
ARM 77	1992	275,000
ARM 78	1992	275,000
ARM 79	1992	275,000
ARM 80	1992	275,000
ARM 81	1992	275,000



- High Performance Computing (HPC)

- Super computers
 - Specialized hardware (NUMA systems)
 - Heterogeneous hardware (FPGAs, GPUs, etc.)
- Precision matters
 - Floating points per second (FLOPS)
- Scientific and analytical use cases
 - OLAP, simulations, forecasts, machine learning, data mining, ...

Both use distributed computing!

- Hyperscale Computing

- Standard computers
 - Fast commodity servers
- Response time, availability and throughput matters
 - X-percentile response time, queries-per-second, ...
- Scalable systems (and analytical) use cases
 - OLTP, web services, application hosting, cloud, data transformation, ...

Big Data Systems

Introduction

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Slide 30

Motivation: "Distributed"

A Rule to Acknowledge

Amdahl's Law

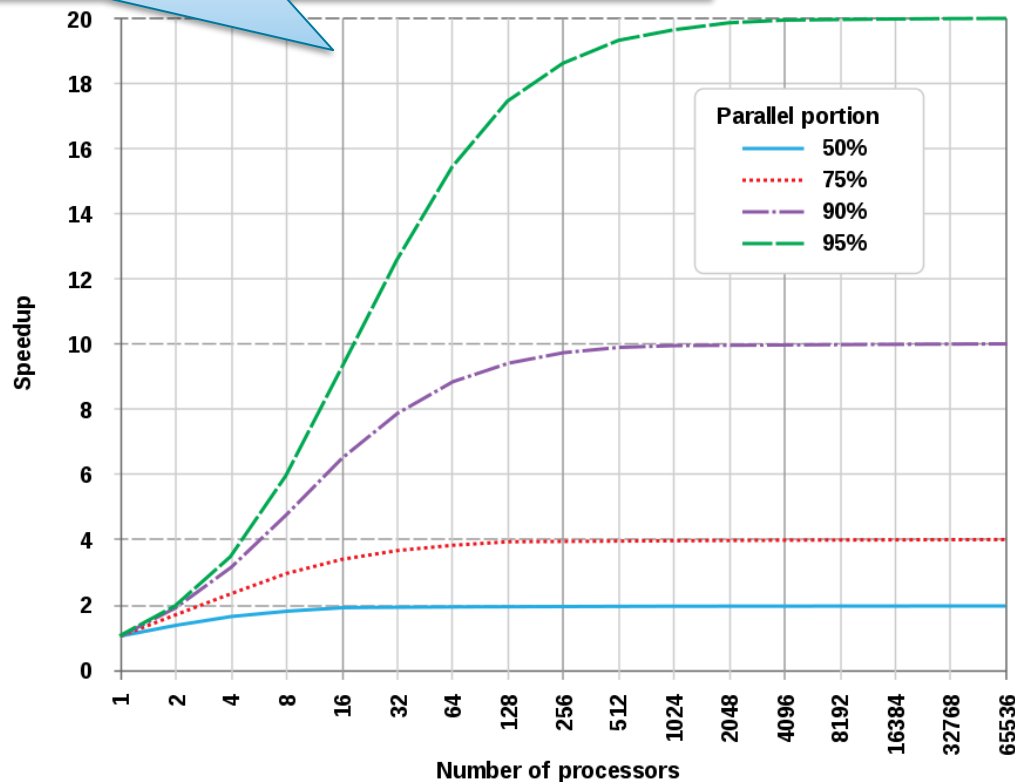
"The speedup of a program using multiple processors for parallel computing is limited by the sequential fraction of the program"

$$\text{Speedup}(s) = \frac{1}{(1 - p) + \frac{p}{s}}$$

s: degree of parallelization (e.g. #cores)

p: percentage of the algorithm that profits from parallelization

Even **distributed parallelization** cannot work around this law!

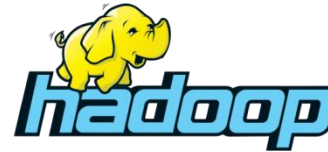


Motivation: "Distributed" New Technologies

Distributed Computing



Distributed Storage



Big Data Systems
Introduction

ThorstenPapenbrock
Slide **32**

THE 2020 MARKET MAP

INFRASTRUCTURE

HAADO ON-PREMISE

cloudera
MAPR
IBM InfoSphere
bluedata
jethro

HAADO IN THE CLOUD

aws
Microsoft Azure
Google Cloud
IBM InfoSphere
treasure data
cazena
CenturyLink

STREAMING / IN-MEMORY

aws
databricks
strim
confluent
GridGain
DataTorrent
dataArtisans
ORACLE
hazelcast
TERRACOTTA

NOSQL DATABASES

aws
Google Cloud
ORACLE
Microsoft Azure
mongoDB
MarkLogic
redshift
GRTA
Alibaba
Couchbase
redislabs
SCYLA

NEWSQL DATABASES

aws
Google Cloud
SAP
Cockroach Labs
MEMSQL
chudata
paradigm

GRAPH DBs

neo4j
Amazon Neptune
ORACLE
ibm
graphio
exasol
snowflake
InfoWorks

MPP DBs - TERADATA

VERTICA
IBM Data Warehouse
Microsoft Azure
Pivotal
snowflake
InfoWorks

CLOUD EDW - aws

Google Cloud
Microsoft Azure
Pivotal
snowflake
InfoWorks

DATA TRANSFORMATION

talend
pentaho
alteryx
trifacta
tomer
Pakata
StreamSets
UNIFI

DATA INTEGRATION

Informatica
snaplogic
segment
redshift
xipity
Import.io
Stitch

DATA GOVERNANCE

Informatica
Salesforce
IBM
collibra
Alation
Waterline
SIKERA

MGMT / MONITORING

New Relic
actio
rubrik
wavefront
dynatrace
collibra
Mogkit
pagerduty
Humint

STORAGE

aws
Google Cloud
Microsoft Azure
PURE STORAGE
alluvio
NetScout
Qumate
cohesity

CLUSTER SVCS - aws

Google Cloud
Microsoft Azure
docus
Keen IO
rainforest
mesosphere
core3

APP DEV

lightbowl
Keen IO
rainforest
mesosphere
core3

CROWD-SOURCING

amazon mechanics
upwork
figure eight
scale
hive

HARDWARE

GPU
arm
MYTHIC
MIDRA
Movidius
WAVE

GPU DBs - kubernetes

IBM
SAP
Hewlett Packard
sas
1010DATA
vmware
TIBCO
TERADATA
ORACLE
NetApp
syncsort

SEARCH

ORACLE
elasticsearch
Elasticsearch
Lucidworks
swiftype
omniscius

LOG ANALYTICS

splunk
sumologic
loggly
logzix

SOCIAL ANALYTICS

hootsuite
sprinklr
NETBASE
synthesio
semarchy
bitly
predata
SimilarWeb

WEB / MOBILE / ANALYTICS

Google Analytics
mixpanel
sumali
RESCI
SIGOPT
granify
custora

ANALYTICS

DATA ANALYST PLATFORMS

Microsoft
pentaho
alteryx
GUUVIS
AYASDI
ATTIVO
Dataquest
interana
ClearStory
Origami
ENDOT
MODE
Bottleneck

DATA SCIENCE PLATFORMS

IBM
KNIME
dataiku
DOMINO
rapidminer
CONTINUUM
ALGORITHMIA
DATAWATCH
ANGELS

SALES

CHORUS
INSIDESALES.COM
CONVERSICA
clari
aviso
tactical
fusemachines
TROOPS

MARKETING - B2B

RADIUS
everest
Lattice
HINTIGO
sense
tubular
datatalk

MARKETING - B2C

zeta
bloomreach
SendGrid
BlueVonder
PERSADO
ACTIONIQ
SAILTHRU
BLUECORE
QUANTIFUNO
mparticle
Amplero
amperity

CUSTOMER SERVICE

MEDALLIA
zendesk
CLARABRIE
Gainsight
NOGATA
DigitalGenius
afiniti
AUTOMAT
frame
msqa
intercom

HUMAN CAPITAL

enlendo
hiq
GISTER
Woxoby
Stella
mya

LEGAL

RAVEL
Seal
JUDICATA
PREVAIL
PREMIER
RESS
casebox

FINANCE

anaplan
ZUORO
ZEPHYRUS
LUMINA
clara
talla
butter
kassito

ENTERPRISE PRODUCTIVITY

slack
box
luminato
difford
clara
talla
butter
kassito

BACK OFFICE AUTOMATION

UiPath
Automation Anywhere
Bluebird
UiPath
Automation Anywhere
Bluebird

SECURITY

TANUIM
CYCLANCE
ILLUMIO
CODE42
Blackops
ANOMALY
Trend Micro
DATASOFT
siftscience
SentinelOne
BitDefender
Avast
Avira
Avast
Avira

ADVERTISING

Apple
criteo
XAd
OpenX
TheTradeDesk
distillery
TAPAR
Clypt

EDUCATION

Blackboard
Canvas LMS
FutureLearn
FutureLearn
FutureLearn

GOVERNMENT

OPENGOV
mark43
FiscalNote
OpenText

FINANCE - LENDING

ondeck
affirm
Kreditcheck
AVANT
INSIGHT
WISDOM

FINANCE - INVESTING

ondeck
affirm
Kreditcheck
AVANT
INSIGHT
WISDOM

REAL ESTATE

Redfin
Opener
VTS
CRIBDI
Zillow

INSURANCE

Insureon
Zillow
Zillow
Zillow

LIFE SCIENCES

Color
Verily
WuXi
ZEPHYRUS
Pilot
Pilot
Pilot

TRANSPORTATION

uber
tesla
clearpath
drive.ai
Pilot
Pilot
Pilot

AGRICULTURE

FAIRFLEX
GRAMER
GRAMER
GRAMER

COMMERCE

Facebook
Pinterest
Pinterest
Pinterest

INDUSTRIAL

GEORX
PREDIX
Gigamon
TACHYUS
TACHYUS

CROSS-INFRASTRUCTURE/ANALYTICS

aws
Google Cloud
Microsoft
IBM
SAP
Hewlett Packard
sas
1010DATA
vmware
TIBCO
TERADATA
ORACLE
NetApp
syncsort

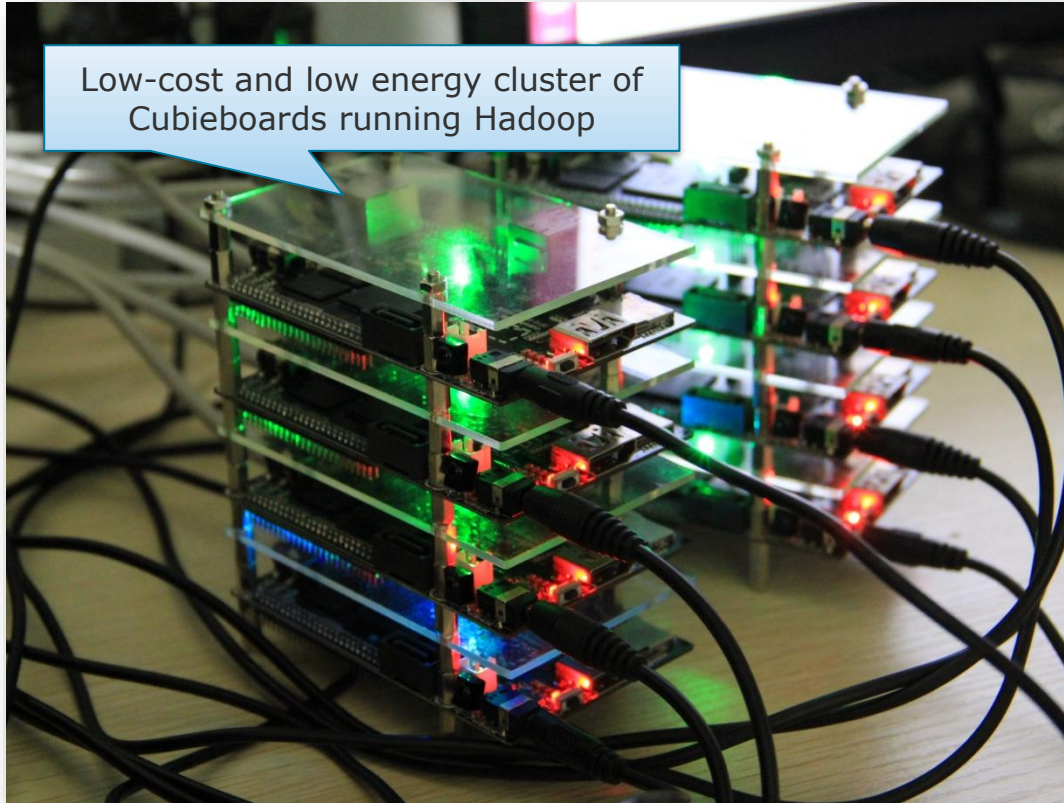
The diagram illustrates the Data Science ecosystem, organized into several key categories:

- FRAMEWORKS:** Includes Hadoop, MapReduce, Hive, Pig, Tez, Flink, Mesos, and Spark.
- QUERY / DATA FLOW:** Includes Spark, SQL, Presto, SLAMDATA, and Google Cloud Dataflow.
- DATA ACCESS:** Includes HDFS, Cassandra, MongoDB, CouchDB, and others.
- COORDINATION:** Includes Apache Zookeeper and others.
- STREAMING:** Includes Flink, Kafka, and Storm.
- START TOOLS:** Includes Python, ScalaLab, and SciPy.
- AI / MACHINE LEARNING / DEEP LEARNING:** Includes TensorFlow, Theano, Caffe, and others.
- SEARCH:** Includes Elasticsearch, Solr, and others.
- LOGGING & MONITORING:** Includes Elasticsearch, Kibana, and others.
- VISUALIZATION:** Includes Tableau, D3.js, and others.
- COLLABORATION:** Includes Jupyter, Anaconda, and others.
- SECURITY:** Includes Apache Ranger and others.
- HEALTH:** Includes Apple, Validic, and others.
- IOT:** Includes GE Digital, Thingworx, and others.
- FINANCIAL & ECONOMIC DATA:** Includes Bloomberg, Thomson Reuters, and others.
- AIR / SPACE / SEA:** Includes Planet, Airbus, and others.
- PEOPLE / ENTITIES:** Includes Axciom, Experian, and others.
- LOCATION INTELLIGENCE:** Includes Foursquare, Esri, and others.
- OTHER:** Includes Qualtrics, Data.gov, and others.
- DATA SERVICES:** Includes Palantir, Uber, and others.
- INCUBATORS & SCHOOLS:** Includes Pluralsight, Galvanize, and others.
- RESEARCH:** Includes Facebook Research, OpenAI, and others.

Motivation: “Distributed” Driving Forces

- **Data volumes increase:**
business data, sensor data, social media data, ...
- **Data analytics gains importance:**
downtime-less, real-time, predictive
- **Parallelization paradigm shifts:**
multi-core and network speeds increase while CPU clock speeds stall
- **Computation resources become more available:**
IaaS, PaaS, SaaS
- **Free and open source software gains popularity:**
setting standards, utilizing external development resources, improving software quality, avoiding vendor locks ...

Motivation: "Distributed" Small and Medium Scale



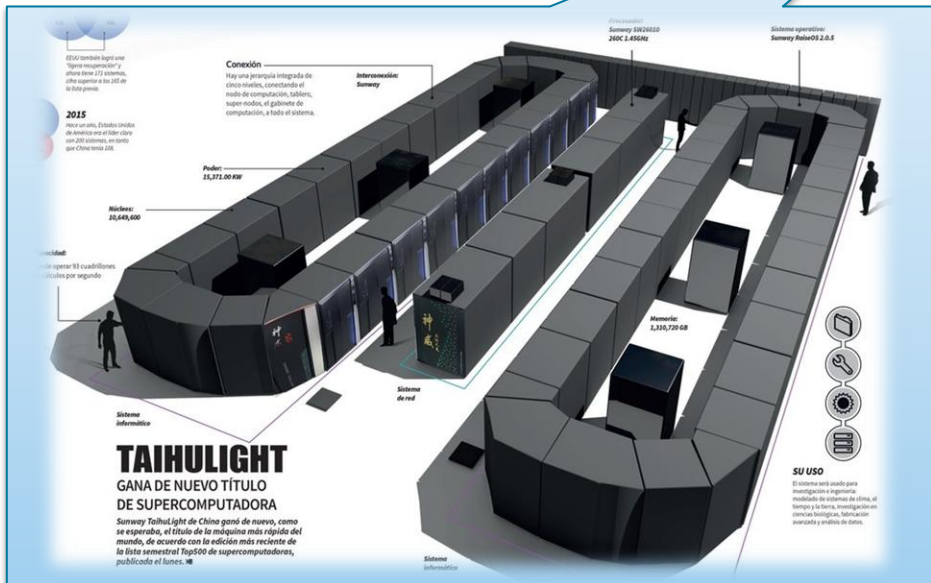
Motivation: "Distributed" Large Scale

A cluster of machines running
Hadoop at Yahoo!



Motivation: "Distributed" Super Large Scale

Top 10 Super Computers 2017



All distributed systems!

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Pow (kW)
1	Sunway TaihuLight - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway , NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
2	Tianhe-2 (MilkyWay-2) - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P , NUDT National Super Computer Center in Guangzhou China	3,120,000	33,862.7	54,902.4	17,808
3	Piz Daint - Cray XC50, Xeon E5-2690v3 12C 2.6GHz, Aries interconnect , NVIDIA Tesla P100 , Cray Inc. Swiss National Supercomputing Centre (SCS) Switzerland	361,760	19,590.0	25,326.3	2,272
4	Titan - Cray XK7, Opteron 6274 16C 2.200GHz, Cray Gemini interconnect, NVIDIA K20x , Cray Inc. DOE/SC/Oak Ridge National Laboratory United States	560,640	17,590.0	27,112.5	8,209
5	Sequoia - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom , IBM DOE/NSA/LLNL United States	1,572,864	17,173.2	20,132.7	7,890
6	Cori - Cray XC40, Intel Xeon Phi 7250 68C 1.4GHz, Aries interconnect , Cray Inc. DOE/SC/LBNL/NERSC United States	622,336	14,014.7	27,880.7	3,939
7	Oakforest-PACS - PRIMERGY CX1640 M1, Intel Xeon Phi 7250 68C 1.4GHz, Intel Omni-Path , Fujitsu Joint Center for Advanced High Performance Computing Japan	556,104	13,554.6	24,913.5	2,719
8	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect , Fujitsu RIKEN Advanced Institute for Computational Science (AICS) Japan	705,024	10,510.0	11,280.4	12,660
9	Mira - BlueGene/Q, Power BQC 16C 1.60GHz, Custom , IBM DOE/SC/Argonne National Laboratory United States	786,432	8,586.6	10,066.3	3,945
10	Trinity - Cray XC40, Xeon E5-2698v3 16C 2.3GHz, Aries interconnect , Cray Inc. DOE/NSA/LANL/SNL United States	301,056	8,100.9	11,078.9	4,233

<https://www.top500.org/lists/2017/06/>

Motivation: "Distributed" Super Large Scale

Top 10 Super Computers 2017



All **distributed** systems!

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<https://www.top500.org/lists/2017/06/>

Motivation: "Distributed" Super Large Scale

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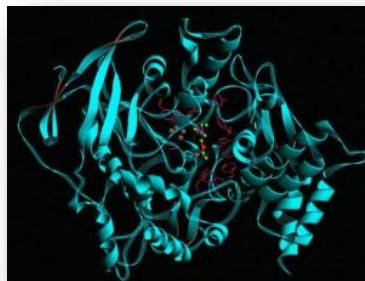
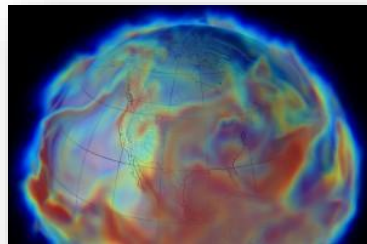
<https://www.top500.org/lists/2017/06/>

Motivation: "Distributed" Super Large Scale

Use cases

- Weather forecasting
- Market analysis
- Crash simulation
- Disaster simulation
- Brute force decryption
- Molecular dynamics modeling
- ...

**Data-intensive analytics
tasks!**



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Rank	System	Cores	max (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	793.40	930.00	13,088
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure United States	2,073,600	561.20	846.84	
6	HPC6 - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461
7	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
8	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	574.84	7,124
9	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107

≈ x1

≈ x2

≈ x3

≈ x1,4

≈ x26

≈ x19

≈ x9

≈ x1,6

≈ x1

≈ x5

≈ x2

≈ x14

≈ x11

≈ x3

≈ x0,6

≈ x3

≈ x1,7

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Introduction

- Distributed Systems
- Lecture Organization
- Motivation “Distributed”
- **Motivation “Data”**
- Motivation “Management”





Data Scientist

The Sexiest Job of the 21st Century

<https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century>





Data Engineer

The 'real' Sexiest Job
of the 21st Century

<https://www.information-age.com/data-engineer-sexiest-job-21st-century-123480578/>



BIG Data & Analytics Software Vendors



Top 10 LinkedIn followers per employee:

1. Dundas
2. Predixion Software
3. iDashboards
4. Chartio
5. CXO-Cockpit
6. Kognitio
7. TARGIT
8. Prophix
9. Alpine Analytics
10. Jedox

A market worth \$122 billion in 2016 with a growth of 11.3% per year!

Excellent job opportunities in many companies!

For a world that created an entire zettabyte (which is exactly 10^{12} GB) of data in the 2010 alone!

Business users (Log-scale)

Shape: Deployment types

On Prem



Cloud (Mobile)

On Prem

Color = Client focus
(Blue = Enterprise ; Orange = SMB ; Green: Mix)

Size = Completeness of vision
(BIG Data, Analytics, CPM, Data Warehousing)

Customers (Log-scale)

Top 5 Self Service BI Vendors:

1. Tableau
2. Microsoft
3. Qlik
4. MicroStrategy
5. GoodData

Top 5 Predictive Analytics Vendors:

1. IBM
2. SAS
3. RapidMiner
4. KNIME
5. Microsoft

Top 10 Business Analytics Vendors:

1. Microsoft
2. MicroStrategy
3. IBM
4. SAP
5. Tableau
6. Qlik
7. Alteryx
8. SAS Institute
9. TIBCO
10. GoodData

Top 5 Data Warehouse Vendors:

1. Teradata
2. SAP
3. IBM
4. Oracle
5. Microsoft

VLDB 2017 Program

International conference
"Very Large Data Bases"

All data processing and
analytics tasks that are
more and more based on
distributed computing.

29 Aug (Tue)			Welcome Messages + Wolfgang Lehner keynote						
8:30-10:00									
10:00-10:30	break								
10:30-12:00			Panel		DB Engines 1	Data Cleaning	Spatial Data Management 1	Graphs and Networks 1	Tutorial: Caching at the Web Scale
12:00-13:30	lunch								
13:30-15:00			Query Processing and Optimization		Stream Processing 1	Data Formats	Privacy and Security	Event Processing	Tutorial: Summarizing Static and Dynamic Big Graphs
15:00-15:30	break								
15:30-16:00			Transactions		Spatial Data management 2	Graphs and Networks 2	Information Integration	Applications	Tutorial: Blockchains and Databases
16:00-16:30				poster reception					
		(Wed)							
8:30-10:00			Endowment Update + VLDB Awards + Michael Franklin keynote						
10:00-10:30	break								
10:30-12:00			High Performance Query Processing		Recommendati ons and Skylines	Approximation Structures	Stream Processing 2		Tutorial: New Trends on Exploratory Methods for Data Analytics
12:00-13:30	lunch								
13:30-15:00			Scalable Storage		Crowdsourcing	Stream Processing 3	Graphs and Networks 3	Social Network Analysis	Tutorial: Geometric Approaches for Top-k Queries
15:00-15:30	break								
15:30-16:00			Concurrency Control		Query Optimization	Specialized Data Management	Graphs and Networks 4	Data Mining and Analytics	Tutorial: Spatial Crowdsourcing: Challenges, Techniques, and Applications
16:00-16:30				poster reception					
16:30-17:00			"VLDB Octoberfest" banquet at the Hofbräuhaus						
		(Thu)							
8:30-10:00			Endowment Awards + Demo Award + Jens Dittrich plenary						
10:30-10:55	break								
10:55-12:00			Transactions and Persistence		Data Access	Data Statistics	Potpouri		
12:00-13:30	lunch								
13:30-15:00			Data Partitioning		Graphs and Networks 5	Visualization	Distributed Systems and Cloud 1	Estimation and Approximation	Tutorial: The Era of Big Spatial Data
15:00-15:30	break								
15:30-17:00			DB Engines 2		Query Processing	Text and Semi- Structured	Spatial Data Management 3	Distributed Systems and Cloud 2	Tutorial: Complex Event Recognition in the Big Data Era
17:00-18:00				poster reception					

Motivation: "Data"

Successful IT Startups

Example: Mobile Motion GmbH



Dubsmash

- An HPI-Startup of 2013
- Founders:
 - Jonas Drüppel, Roland Grenke, Daniel Taschik

November 19, 2014:

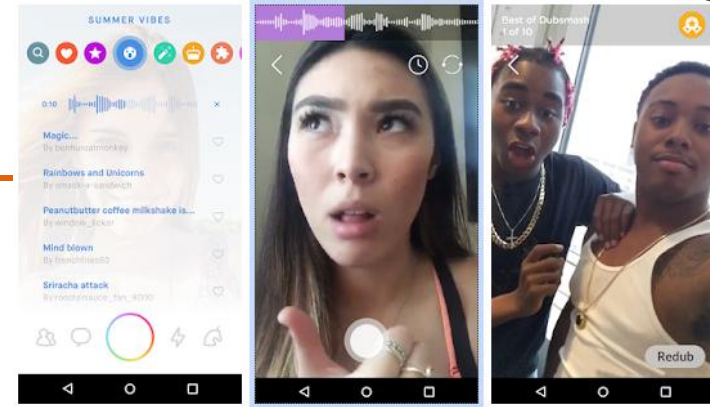
- **Launch** of the Dubsmash app

November 26, 2014:

- Dubsmash reached the **number one** downloaded app in Germany

June 1, 2015:

- Dubsmash had been downloaded over **50 million times in 192 countries**



Big Data Systems

Introduction

ThorstenPapenbrock

Slide 47



HPI Start-ups



- 23.04.2016 | HPI Startup Talks: Lecture 4
→ mehr
- 15.01.2016 | Startup Talk@HPI: Lecture 3
We will resume our Startup Talk@HPI-Series in 2016 with a talk by jochem Engert, Co-Founder & CEO ... → mehr
- 11.10.2017 | Startup Talk@HPI: Lecture 2
On the second event of our Startup Talk@HPI-Series startup lawyer Andre Eggert and serial ... → mehr
- 24.11.2017 | HPI Startup Bootcamp 2017
Dear Entrepreneurship Enthusiasts, does gaining first-hand entrepreneurial experience with real ... → mehr
- 09.11.2017 | Startup Talk@HPI: Lecture 1
Learn 'How to go from an idea to a GREAT business plan. Learnings from GetHourGuide' in our first ... → mehr
- 30.09.2016 | Gründerwettbewerb - Digitale Innovationen
Courtroom für Deutschlands beste ICT-Könnchen: 10 Top-Ins zum Bewerbungsfinale des ... → mehr
- 29.04.2016 | HPI Start-up Spotlight 2016
Beim 2. HPI Start-up Spotlight am 29.4.2016 ab 17:00 Uhr berichteten HPI-Alumni und andere Jurys ... → mehr
- 24.04.2016 | Bewerbungsschluss Businessplan-Wettbewerb 2016
Alle zwei Jahre veranstaltet das Hasso-Plattner-Institut (HPI) einen Businessplan-Wettbewerb für ... → mehr
- 11.04.2016 | Kick-off HPI-Businessplan-Wettbewerb 2016
Alle zwei Jahre veranstaltet das Hasso-Plattner-Institut (HPI) einen Businessplan-Wettbewerb für ... → mehr
- 20.12.2015 | New York ruft: Start-up-Stipendium des Media Residency Program
Für 8 Monate nach New York: Stipendium werden für zwei Stipendiaten für Gründer, Künstler und ... → mehr
- 14.12.2015 | Senior Coaching Service-Wettbewerb
Bis zum 12. Mai findet an der Universität Potsdam der Businessplan-Coaching-Wettbewerb des ... → mehr
- 01.12.2015 | Gründer-Spendenaktion "plug & play"
Am 1. Dezember 2015 findet das vom Potsdamer Transfer gemeinnützigen mit Axel Springer Plug and Play ... → mehr
- 20.11.2015 | Gründerwettbewerb 2015
Der Wettbewerb wird vom 10. - 22. November 2015 am HPI unter Federführung ... → mehr
- 08.11.2015 | F&E Lab
Das Lab ist eine interdisziplinäre Nachwuchswissenschaftler aus ... → mehr
- 24.10.2015 | Der Entrepren-Garten
→ mehr
- 23.09.2015 | Experience
Dem 'Totale experiment' die ... → mehr
- 23.08.2015 | R&D-Rap Challenge
Entwerfen einer Idee zum Connected Car im interdisziplinären Studententeams ... → mehr
- 15.08.2015 | Startup Schuber Innovation Slam
Wettbewerb für Jungunternehmen, den speziell Talente der Marketingbranche angesucht ... → mehr
- 30.06.2015 | SAP Startup Forum
Das SAP Startup Forum Programm unterstützt erfolgreiche vernetzte Startups im Bereich Big Data ... → mehr
- 16.05.2015 | 12. Idea Pitch-Night: From Ideas to Implementation - DT & IT Meet Entrepreneurship
Der D'School Kick-It-and-Design Thinking-Studenten ein, um Euch und Euer innovativen Ideen ... → mehr
- 10.05.2015 | Start-up Kit im MIZ-Bahnhof
Beim „Phonetechnik“ erhalten Ihr das nötige Expertenwissen und geeignete Entwicklungsalgorithmen ... → mehr
- 29.05.2015 | Idea Pitch-Night: From Ideas to Implementation - DT & IT Meet Entrepreneurship

Many further HPI Startups!

Motivation: "Data"

Successful IT Startups

Successful IT-Startups in recent years are masters of data:

1. **AirBnB**
2. **Instagram**
3. **Pinterest**
4. **Angry Birds**
5. **Linkedin**
6. **Uber**
7. **Snapchat**
8. **WhatsApp**
9. **Twitter**
10. **Facebook**
11. ...

Peta- to Exabytes of ...

- profile data (names, addresses, friends, ...)
- content data (images, videos, messages, ...)
- event data (logins, interactions, games, ...)
- ...

Challenged with ...

- streaming
- persistence
- analytics
- load-balancing
- ...

Big Data Systems

Introduction

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Slide 49

Introduction

- Distributed Systems
- Lecture Organization
- Motivation “Distributed”
- Motivation “Data”
- **Motivation “Management”**

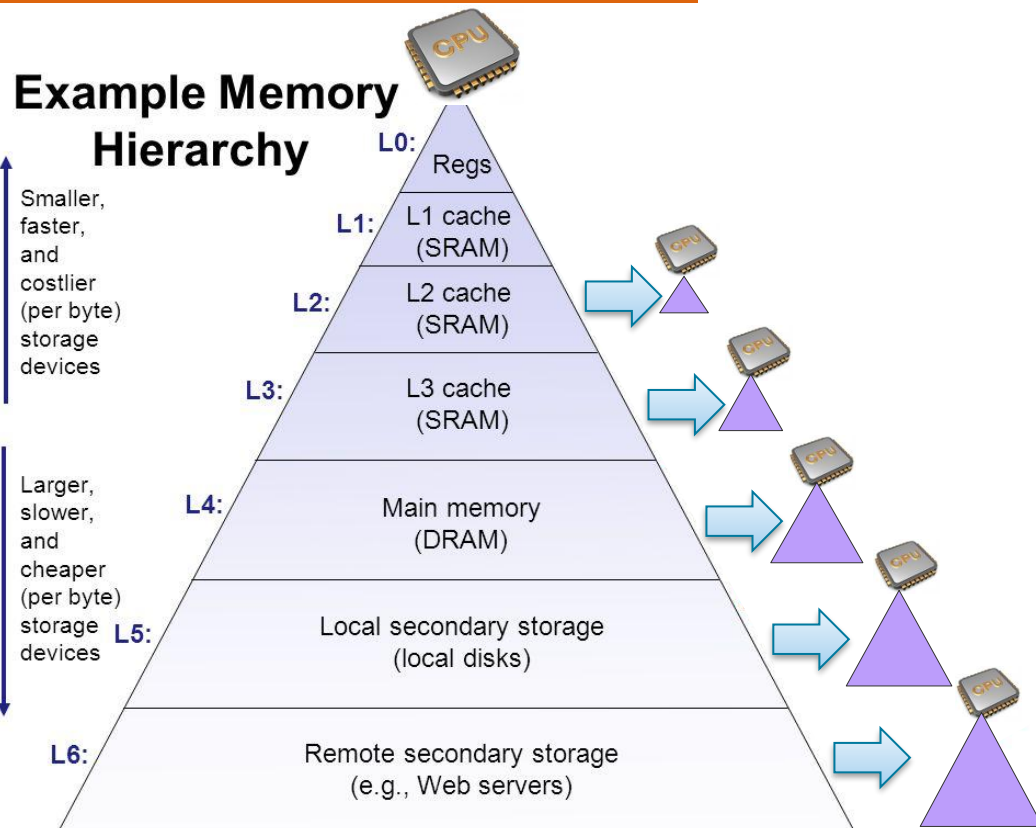


Motivation: "Management"

Rethinking Data Management

Data is distributed and replicated!

- Data needs to reach a processor to be computed.
- Processor memory is very small but data is usually large.
- Data is stored distributed and replicated in memory hierarchies.
- Data needs to be fetched, i.e., copied to a processor before it can be computed.
- Data needs to be flushed, i.e., copied to higher memory levels to become visible to other processors.



Motivation: "Management"

Rethinking Data Management

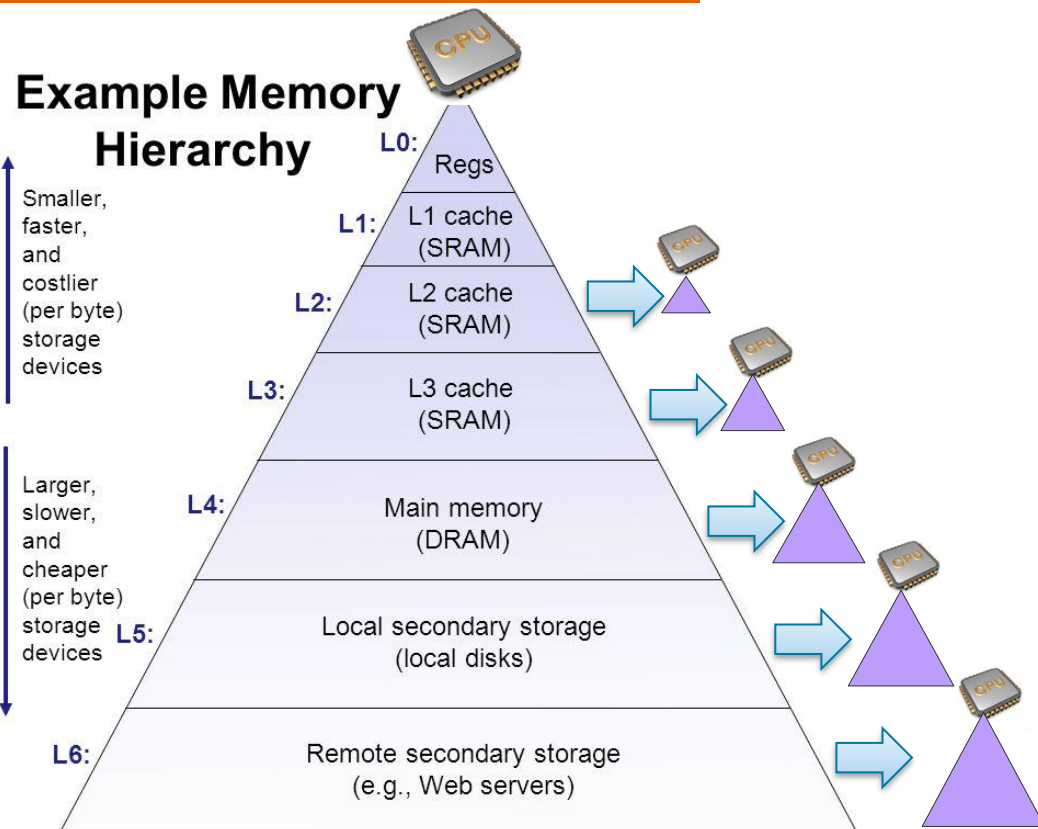
Moving data costs magnitudes more time and energy than computing data!

- Copying data costs time and energy.
- Stalled processors during data copying consume energy.

Operation	Operation Energy Cost (nJ)	Equivalent ADD
ADD	0.64	-
L1->REG	1.11	1.8x
L2->REG	2.21	3.5x
L3->REG	9.80	15.4x
MEM->REG	63.64	99.7x
Stall	1.43	-
Prefetching	65.08	-

<https://hpc.pnl.gov/modsim/2014/Presentations/Kestor.pdf>

➤ **Push computation to the data
not data to the computation!**



Motivation: "Management"

Rethinking Data Management

Moving data costs magnitudes more time and energy than computing data!

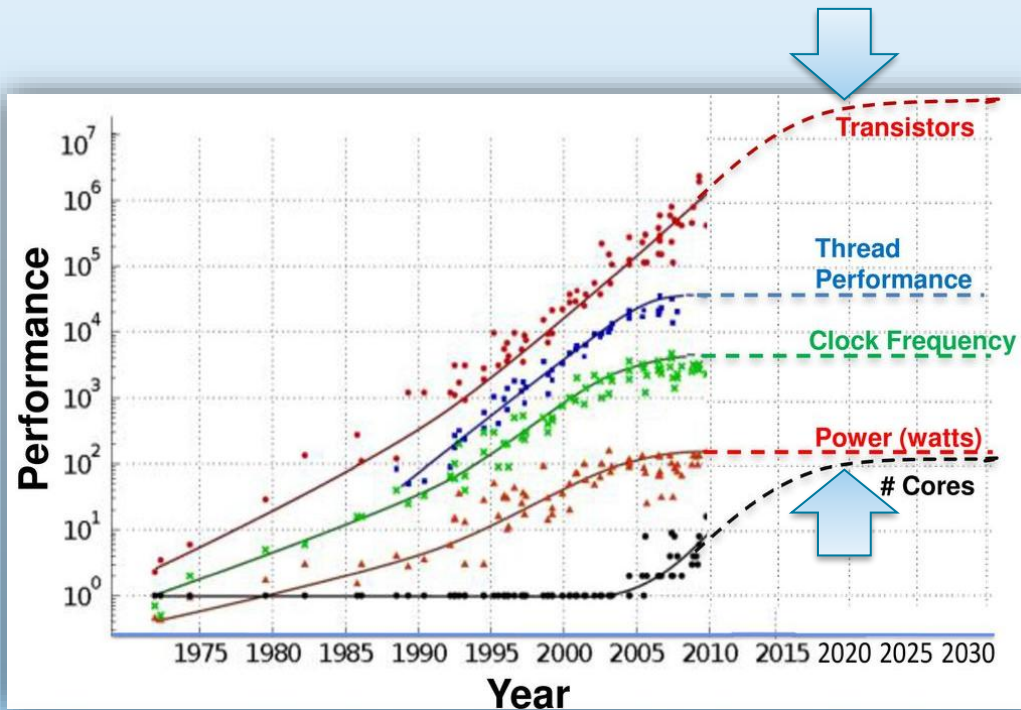
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- **Push computation to the data
not data to the computation!**

Why energy is a concern:



Motivation: "Management"

Rethinking Data Management

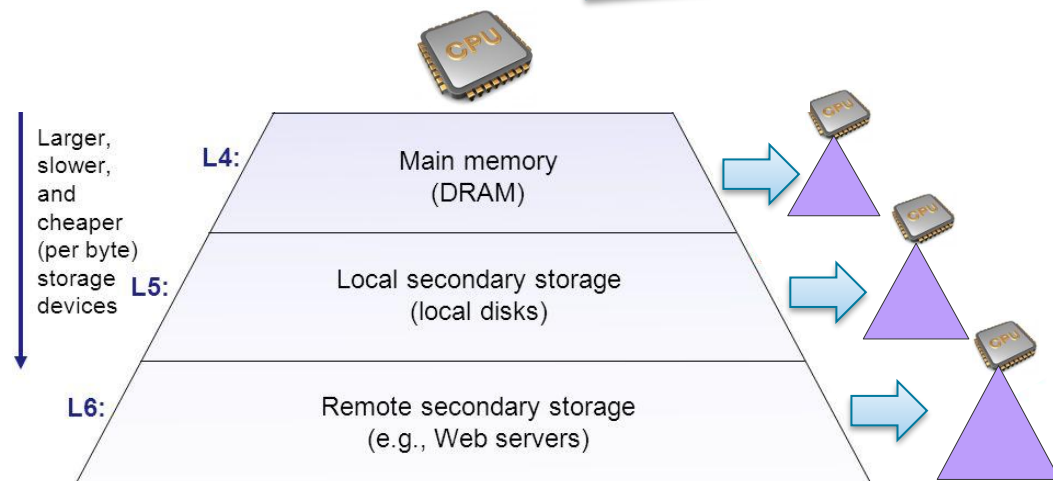
➤ Data engineers and data scientists need to be good data manager!

- Data encoding
- Data transmission
- Data replication
- Data partitioning
- Data consistency management
- Load scheduling
- Load balancing

We do not consider L0-L3 in this lecture, but this is super relevant for High Performance Computing!

I recommend:

https://www.youtube.com/watch?v=3PjNgRWmv90&list=LLbLaqsrSDDURdv_ZV75-AMQ&index=6&t=0s





Philipps

Universität
Marburg

Domain Expertise (e.g., Industry 4.0, Medicine, Physics, Engineering, Energy, Logistics)

Mathematical Programming

Relational Algebra / SQL

Linear Algebra
Statistics

Data Warehouse/OLAP
NF² / XQuery

RDF / SparQL

Information Integration

Information Extraction

Visual Analytics

Privacy

Memory Management

Parallelization

Scalability

Memory Hierarchy

Fault Tolerance

Security

Data Analysis Languages

Query Optimization

Real-Time

Indexing

Control Flow

Data Flow

Legal Aspects

Business Models

Curse of Dimensionality

Iterative Algorithms

Decoupling

Convergence

Data Obfuscation

Sketches

Predictive Analytics

Regression

Monte Carlo

Active Sampling

Error Estimation

Machine Learning

Stochastic Gradient Descent

Signal Processing

Graph Mining

Text Mining

Statistics

Domain
Knowledge

Data
Science

DA
Data Analytics

DM
Data Management

Motivation: “Management” Data Management

Data Analytics

“The ability to effectively
extract and calculate
various kinds of information from data!”

- Structural information
- Explicit information
- Implicit/derived information

Data Management

“The ability to efficiently
read, transform, and store
large amounts of data!”

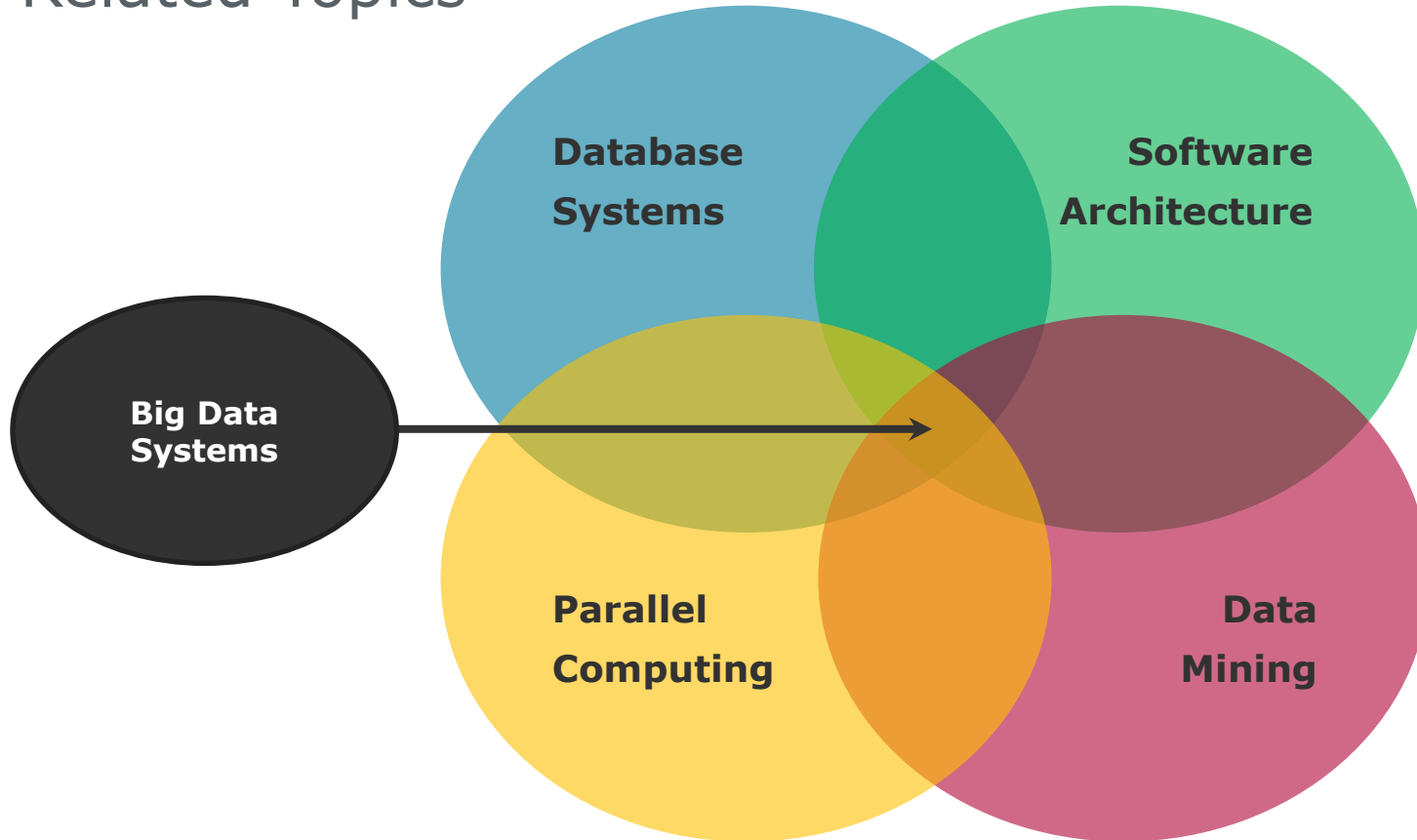
- Static (block) data
- Volatile (streaming) data

Big Data Systems

Introduction

Motivation: "Management"

Related Topics

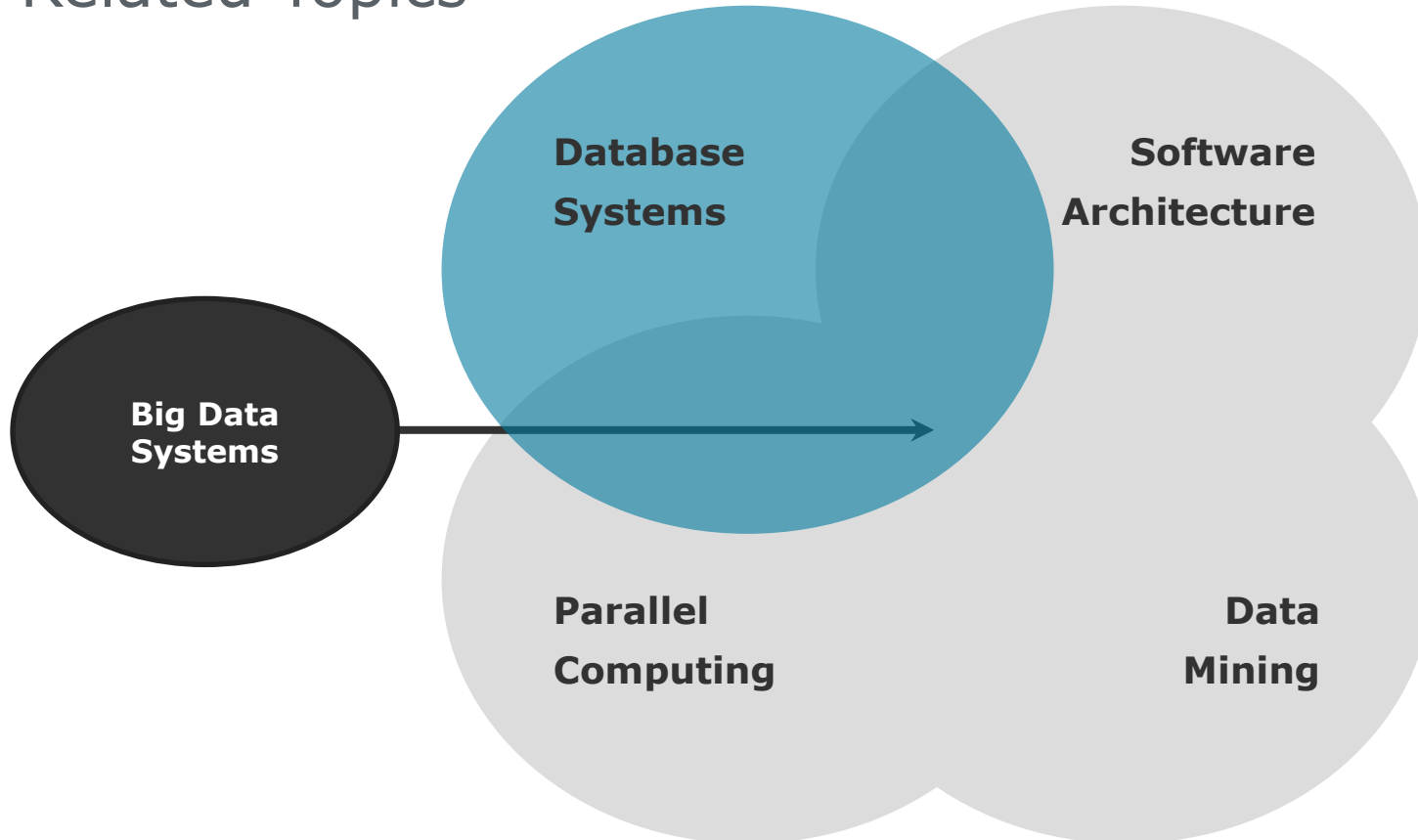


Big Data Systems
Introduction

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Slide **57**

Motivation: "Management"

Related Topics



Big Data Systems
Introduction

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Slide **58**

Motivation: “Management” Database Systems

Touch points

- Data models, query languages, and consistency guarantees
- Distributed storage and retrieval of data
- Index structures and NoSQL data
- Transactions and query optimization

Not in this lecture

- Physical data storage
- Foundations on transaction management and logging

More focused lectures/seminars

- Datenbanksysteme (Prof. Seeger, Prof. Papenbrock)

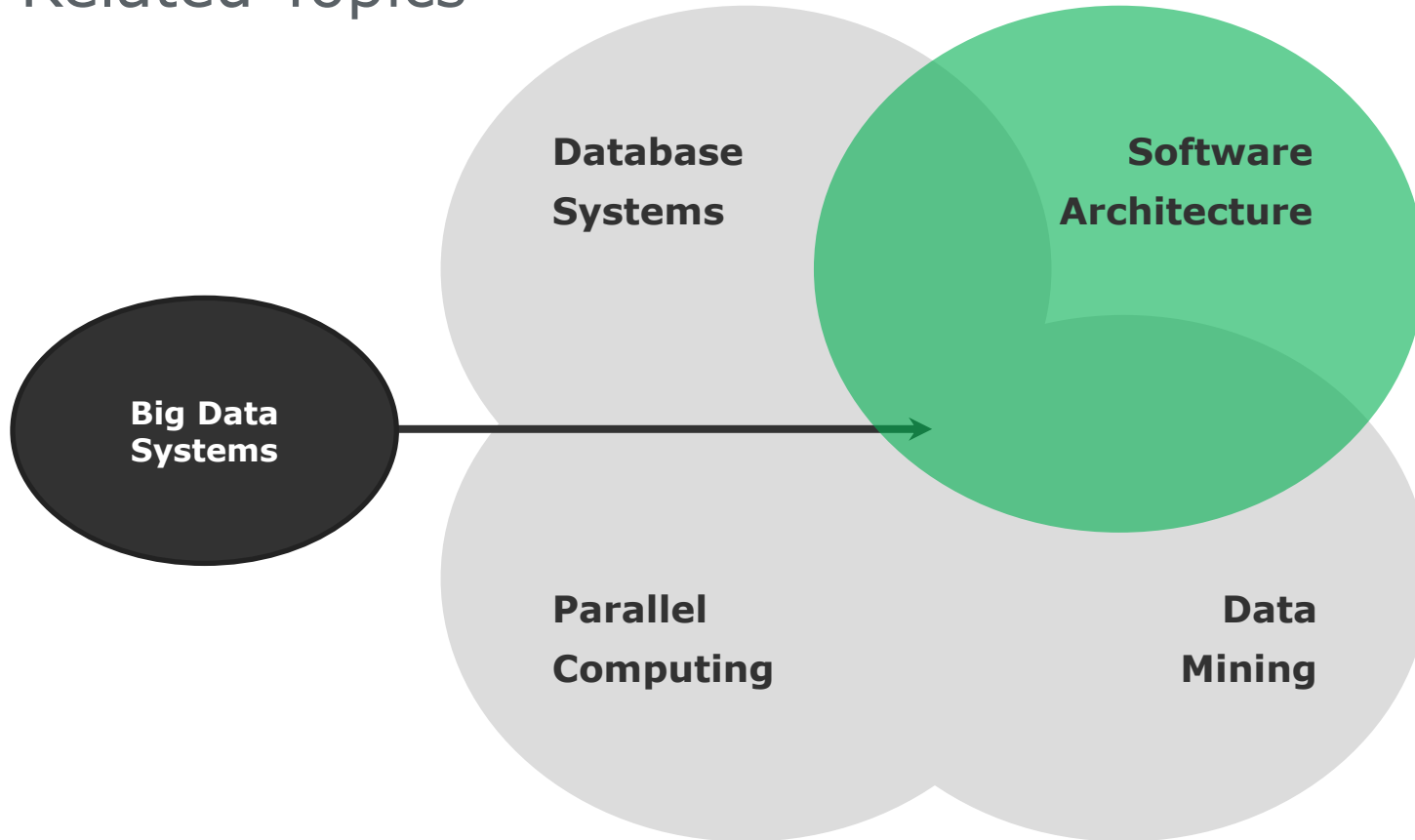
Big Data Systems

Introduction

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Slide **59**

Motivation: "Management"

Related Topics



Motivation: “Management” Software Architectures

Touch points

- Requirements, design, and architecture of distributed systems
- Pros and cons of different technologies for distributed systems

Not in this lecture

- Non-distributed systems
- Agile software development techniques
- Basic software patterns

More focused lectures

- Softwaretechnik (Prof. Tüntzer)
- Softwareentwicklungswerkzeuge für datenwissenschaftliche Anwendungen (Prof. Bockisch)

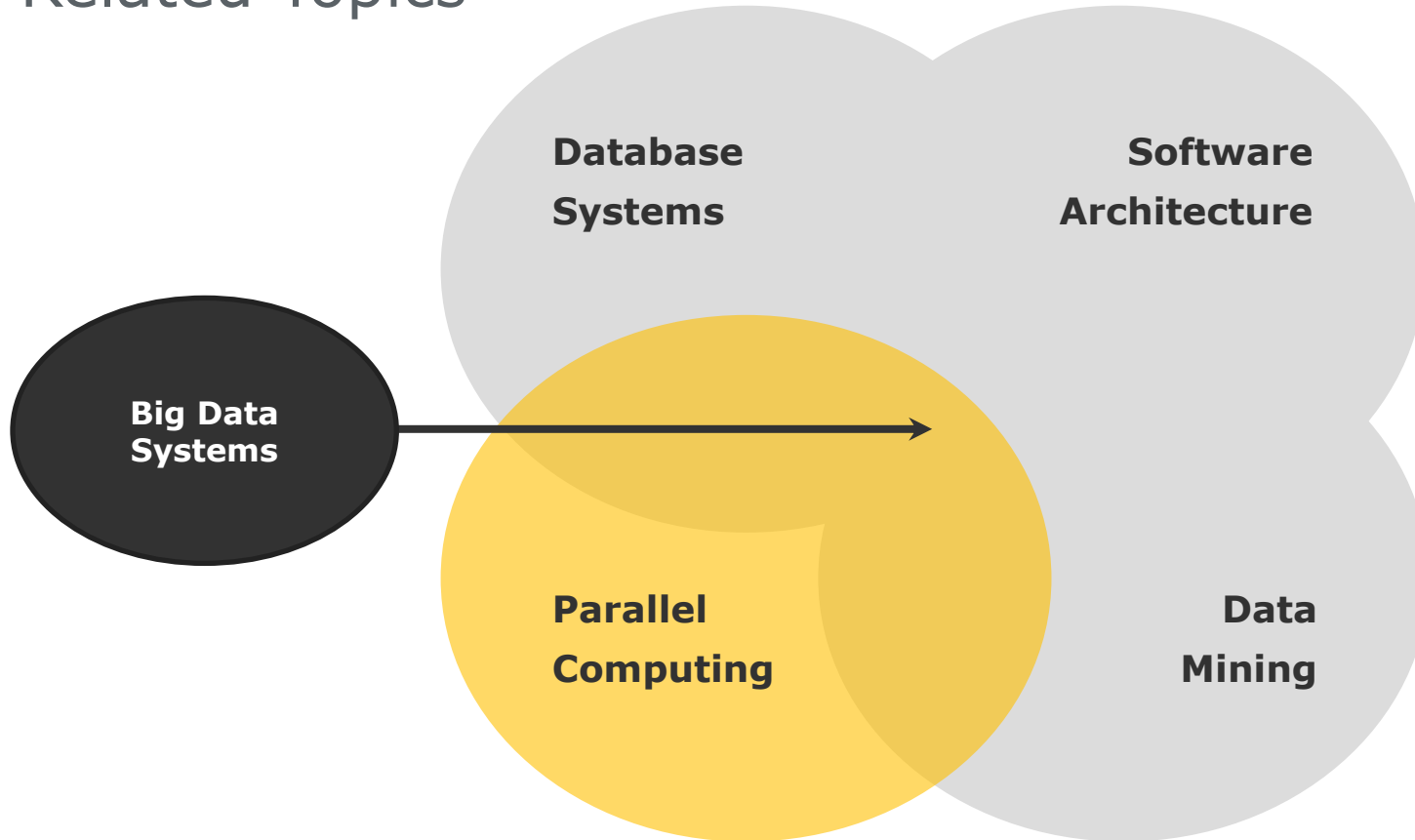
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Slide **61**

Motivation: "Management"

Related Topics



Big Data Systems
Introduction

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Slide **62**

Motivation: “Management” Parallel Computing

Touch points

- Distributed data storage concepts
- Distributed programming models, e.g., actor programming and MapReduce

Not in this lecture

- Parallel, non-distributed programming languages, e.g., CUDA or OpenMP
- Core parallel computing concepts, e.g., scheduling or shared memory
- Processor architectures, cache hierarchies, GPU programming, ...

More focused lectures

- (?) Systemsoftware und Rechnerkommunikation (Prof. Freisleben)
- (?) Betriebssysteme (Prof. Freisleben)

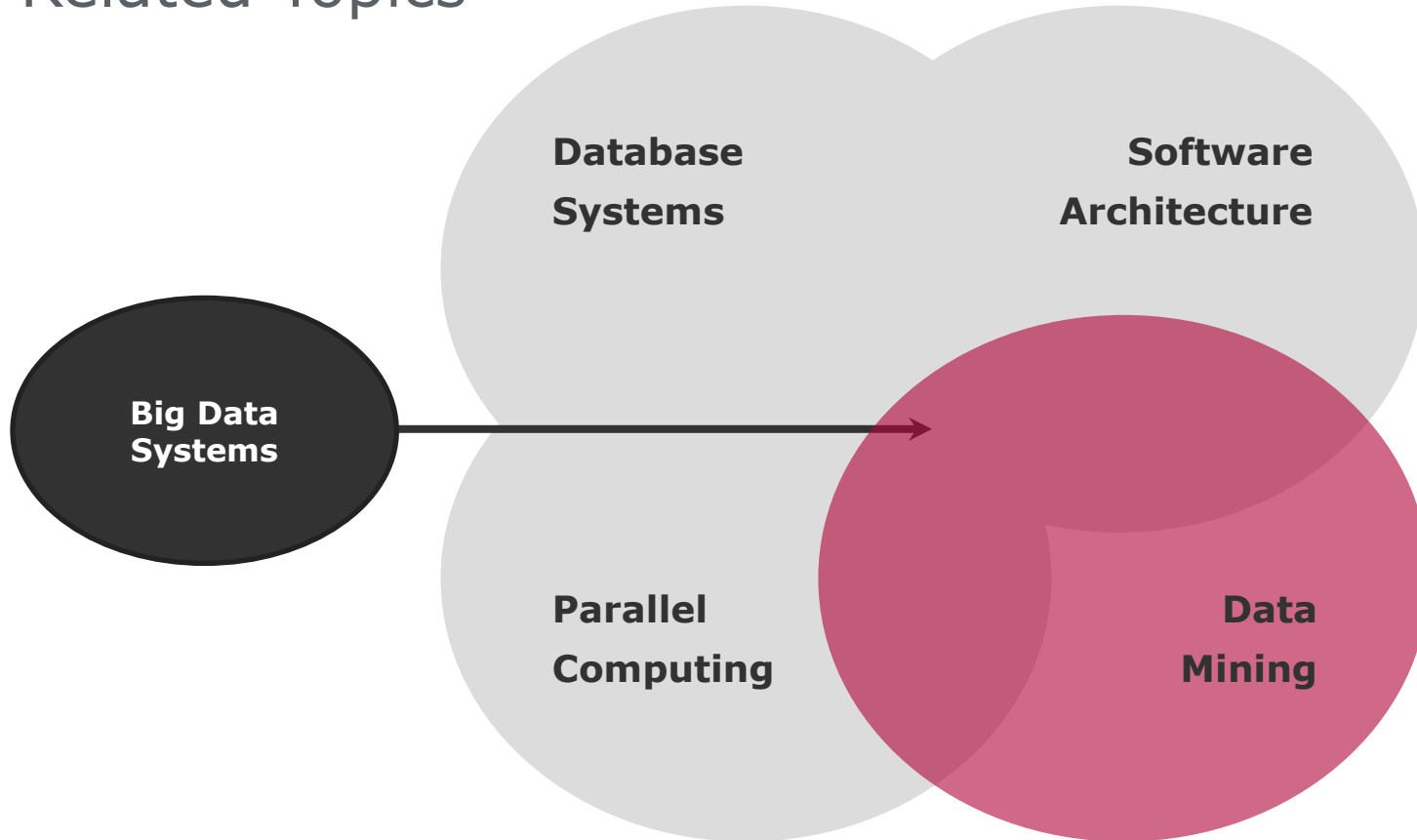
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Slide **63**

Motivation: "Management"

Related Topics



Big Data Systems
Introduction

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Slide **64**

Motivation: “Management” Data Mining

Touch points

- Efficient data querying and aggregation queries
- Data engineering for data analysis
- Scalable computing

Not in this lecture

- Machine learning, e.g., neuronal networks, (un)supervised learning, or Bayesian classification
- Statistics, linear algebra, and most sophisticated mining algorithms

More focused lectures/seminars

- Maschinelles Lernen (Prof. Heider), Data Science in Biomedicine (Prof. Heider), Web & Social Media Mining (Prof. Flek), ...

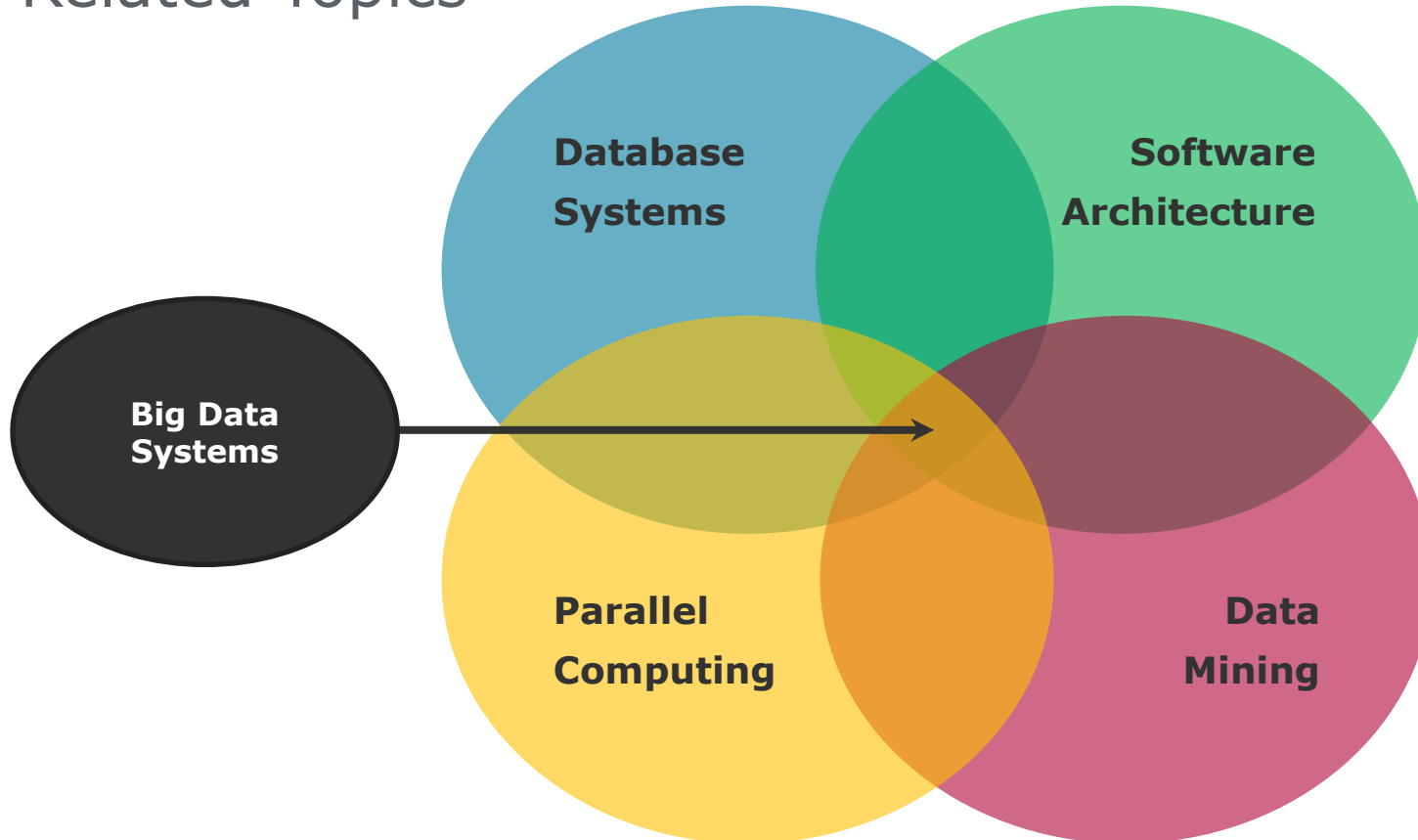
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Motivation: "Management"

Related Topics



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Introduction

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Slide **66**

Motivation: “Management”

Lecture Goals

Sorting the buzzwords

- NoSQL, Big Data, OLAP, Web-scale, ACID, Sharding, MapReduce, Scale-out...

Understanding distributed systems

- You know how state-of-the-art distributed systems work.
- You know core technologies and techniques to solve distributed challenges.
- You know the advantages and disadvantages of important systems.
- You know how to handle data in distributed settings.



Exercising in distributed data management and analytics

- You can implement distributed algorithms and applications.
- You can solve problems that arise in distributed setups.
- You can write data-parallel and task-parallel applications.

Big Data Systems

Introduction

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“Dark Magic”

- With **distributed computing** we can utilize incredible amounts of compute power!
 - At the cost of harder programming (e.g. fault tolerance, testing and protocols)
 - At the cost of additional energy (e.g. communication and redundancy)
- Efficient, fault resistant code matters all the more, because inefficiency and failures scale, too!



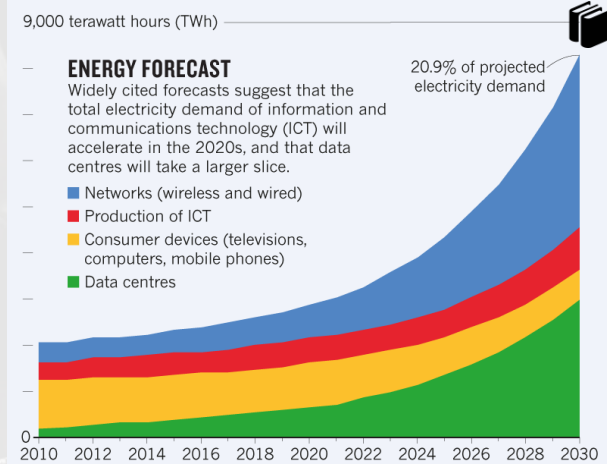
"Dark Magic"

- "Around 10% of the world's total electricity consumption is being used by the internet."
Swedish KTH
<https://www.insidescandinavianbusiness.com/article.php?id=356>
<https://www.sciencedirect.com/science/article/pii/S2214629618301051>
- "The Internet's data centers alone may already have the same CO2 footprint as global air travel."
Global e-Sustainability Initiative
<https://internethealthreport.org/2018/the-internet-uses-more-electricity-than/>
- "Data centres [...] consume about 3% of the global electricity supply [...] accounting for about 2% of total greenhouse gas emissions" in 2016.

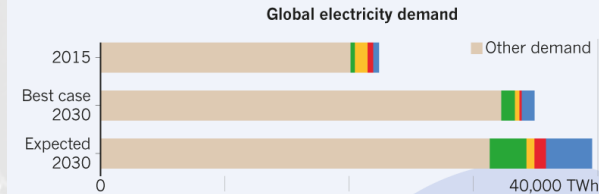
Independent

<https://www.independent.co.uk/environment/global-warming-data-centres-to-consume-three-times-as-much-energy-in-next-decade-experts-warn-a6830086.html>

<https://www.nature.com/articles/d41586-018-06610-y>



The chart above is an 'expected case' projection from Anders Andrae, a specialist in sustainable ICT. In his 'best case' scenario, ICT grows to only 8% of total electricity demand by 2030, rather than to 21%.



INTERNET EXPLOSION

Internet traffic* is growing exponentially, and reached more than a zettabyte (ZB, 1×10^{21} bytes) in 2017.

1987 2 TB†
1997 60 PB
2007 50 EB
2017 1.1 ZB

*Traffic to and from data centres.

†TB, terabyte (10^{12} bytes); PB, petabyte (10^{15} bytes); EB, exabyte (10^{18} bytes).

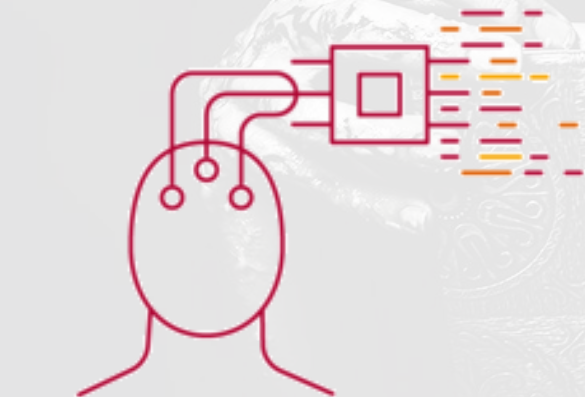
“Dark Magic”

- “In kürzester Zeit wird die Digitalisierung zum Klimaproblem Nummer eins werden.”

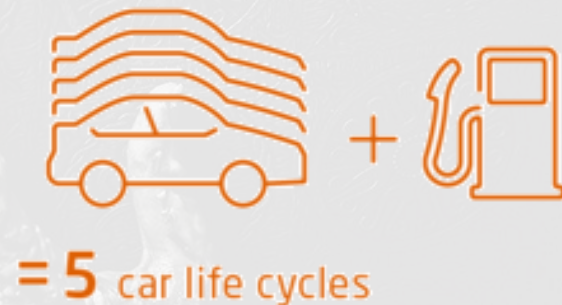
HPI, clean-IT Initiative

<https://hpi.de/open-campus/hpi-initiativen/clean-it-initiative.html>

CO2-footprint



1 AI model training



<https://www.aclweb.org/anthology/P19-1355.pdf>

https://theoutline.com/post/8186/artificial-intelligence-destroy-environment?utm_campaign=Artificial%2BIntelligence%2BWeekly&utm_medium=email&utm_source=Artificial_Intelligence_Weekly_131&zd=1&zi=cayumttz



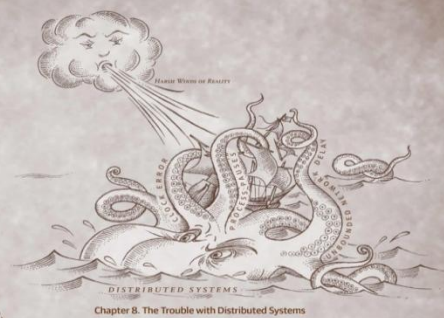
Chapter 1. Reliable, Scalable, and Maintainable Applications



Chapter 2. Data Models and Query Languages



Chapter 3. Storage and Retrieval



Chapter 8. The Trouble with Distributed Systems



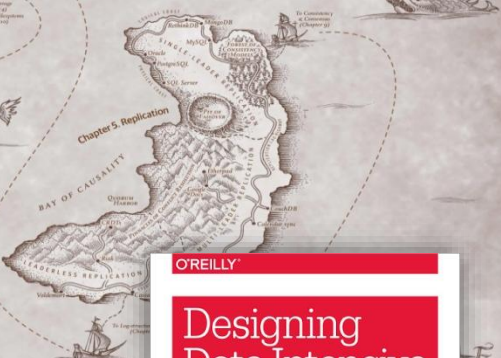
Chapter 7. Transactions



Chapter 4. Encoding and Evolution



Chapter 9. Consistency and Consensus



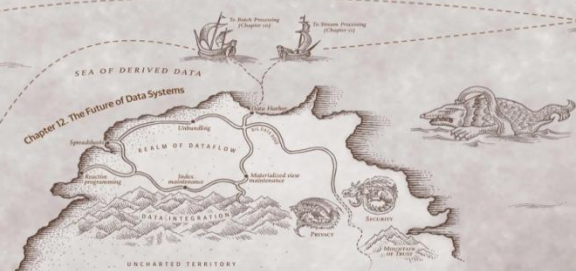
Chapter 5. Replication



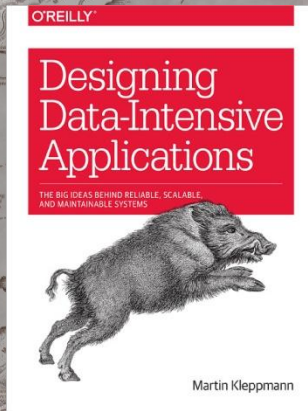
Chapter 6. Partitioning



Chapter 10. Batch Processing



Chapter 12. The Future of Data Systems



Martin Kleppmann